No False Green: Evidence Sufficiency and Unsupported Decisions in Language Models

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Abstract. A system that marks a decision satisfied claims that its evidence establishes it. We study when language models make that claim without support, giving **false greens**, under rules that state which records decide each question, and what removes them. Every label is checked by independently written programs. In employer immigration compliance, adding **proxy columns** (fields beside a missing required record, such as an onboarding date beside a missing Form I-9) raised the false-green rate by 0.427 under a pass/fail dashboard schema and by 0.217 when abstention was offered, in a preregistered study of seven models from five developers (0.244 and 0.237 with firms weighted equally; exact permutation p ≤ 0.0013, Holm). On 120 real public LCA records, a roster's pay fields raised them by 0.236 with abstention offered, and the salary-of-record column alone induced false greens in three of five models. Removing the abstaining verdicts (+0.384) or a one-sentence warning (+0.093) raised false greens and an evidence-first prompt lowered them (−0.074), but no menu, prompt or explicit decision contract reliably removed them, and declared tests of four explanations found only a partial one. What sharply reduced it was separating reading from deciding: the model extracts each fact with its source, and a solver applies the rule. Across three workflows, averaged over five models, this cut unsupported positives from 0.068 to 0.002 and raised correct decisions from 0.818 to 0.967. In a declared, powered transfer test on six held-out rules and 30 fresh record sets, rendered in six layouts by three designers who worked independently of us and of the baselines, it lowered unsupported positives by 0.094 against the strongest prompt, in all 30, and made 0.544 more correct decisions than a rule-based parser developed on six other layouts and 0.172 more than parsers the models wrote themselves; for two of five models it cost some safety. On 600 real public records in their publishers' formats it again cut unsupported positives against the prompt (by 0.094) and raised correct decisions (by 0.171), in every packet; a model that wrote a parser, given the pairs to answer and the diagnostics of failed runs, showed no detectable difference in accuracy (0.972 against 0.964), with opposite differences for different models, and made no unsupported positive in 6,000 evaluated pairs. A first run of that study had truncated some records for two of its arms; we report the correction. Keeping the decision itself out of the model sharply reduces unsupported decisions on these contracts; it does not guarantee their elimination, since a reader can still misread or suppress a fact. All data, prompts and scored replies are released (16 capped Study 15 completions that were re-requested are counted but not kept, and Study 15's defective first run kept only its final code-generation programs), and a script recomputes every number.

1 Introduction

A system that marks a decision as satisfied claims more than a probable answer: it claims the evidence it was given establishes that answer. A green cell on a compliance dashboard asserts that an obligation was checked against the record and found met. When the record was not there, the system can say so or render the cell green anyway, and a mostly green dashboard looks like a working product. We call the second choice a **false green**: an unsupported positive decision. This paper asks when language models make it, given rules that state which evidence decides each question, and what reduces it.

A false green is not the mirror image of a false alarm. A false alarm costs review time and is corrected by the review it triggers; a false green triggers nothing, and the review it suppresses is the only thing that would surface it. In immigration compliance the cost is concrete: Form I-9 paperwork violations are penalised per form (8 C.F.R. 274a.10(b)(2)), and violations of a Labor Condition Application (LCA), the wage attestation behind an H-1B petition, per violation (20 C.F.R. 655.810(b)). A gap found before an inspection can often be cured; found during one, it becomes a finding. The wage gap is not hypothetical: in fiscal year 2025 the Department of Labor's Wage and Hour Division found H-1B violations in 74 compliance actions and recovered $3,306,457 in back wages for 152 workers (U.S. Department of Labor, 2026), each a case in which wages an LCA required were not fully paid. The figures show that the obligation is enforced with consequences; they do not show what records those employers kept or what a compliance system would have reported.

Pass/fail evaluation cannot see it. An evaluation with no category for "the evidence was not there" never penalises a system that answers `pass` whenever evidence is absent. Recent benchmarks score overclaiming under insufficient evidence for agents (Solozobov, 2026; Tallam, 2026), and legal and tabular benchmarks include a "not enough information" answer (§2); none scores regulatory verdicts over a third party's operational records when the required record is missing, where declining to decide is the correct answer for about half the items.

Rule engines make it too. The commercial engine our rulebook derives from (§4; see Competing interests) had the same substitution written into deterministic code. Reviews in June and July 2026 found that two I-9 evaluators skipped employees whose records lacked the deciding dates, dropping them from monitoring and leaving the dashboard green; that a wage rule let a salary of record decide whether

wages were paid; and that the required-wage rule passed without the employer's actual-wage benchmark. Each now returns `needs_evidence` (Appendix H). The engine was written with AI coding assistance and the defects were found in AI-assisted review, so this shows that the shortcut is easy to build into a system, not how human engineers behave.

We built an instrument for this error (Figure 1) and report four findings. **Separating reading from deciding sharply reduces it, and transfers.** A model that only extracts each fact with the record it came from, followed by a deterministic solver, sharply reduced unsupported decisions while raising correct ones under explicit decision contracts in three workflows, which no prompt or record-presence gate matched. On six rules and three record layouts held out from the method's design, the layouts written by a separate author, it kept that safety and made more correct decisions than a strong rule-based parser that was exact on the layouts it was written for (§6.12). A powered test on 30 fresh record sets in six layouts from three independent designers confirmed this against an improved parser and against parsers the models wrote themselves, while showing that for two of five models the gain costs some safety (§6.13). On real public records in their native formats, extraction and a model-written parser, given the pairs to answer and the diagnostics of failed runs, showed no detectable difference in pooled accuracy, with large differences in both directions for particular models, and the parser made no unsupported decision (§6.14). In every successful design, the decision itself belonged to a program. **Proxy columns are a large cause:** planting columns that resemble required evidence, with the prompt unchanged, raises the false-green rate by 0.217 to 0.427. Controls show that it is not the number of added columns; for Google's models it is mostly their names; the two Gemini models take a proxy that suggests compliance as evidence while discounting one that suggests a breach, and two other developers' models respond to the values and substitute in both directions. **Of the output schema, removing the verdicts for "could not tell" matters most** for all five models, and removing a one-sentence warning that a missing record is not a record in order also raises false greens. **High recall does not guarantee safety:** in Study 3, none of the four models that detected at least 0.92 of violations came close to zero false greens on proxy-column firms, although on real LCA records Gemini 3.1 Pro both detected every underpayment and made no false green (§6.5).

Each ingredient has precedents (§2): answers for insufficient evidence, directional overclaiming, distracting context and answer-menu effects. Our contribution is their combination where it has consequences: a four-verdict task over a third party's operational records with a stated primary-record standard (§3); generators that fix labels before data, with every label re-derived by separately written checkers and every "insufficient evidence" label that turns on missing values certified by two opposite completions, conflict labels by agreement (§4); a preregistered study and declared follow-ups that separate the proxy effect into name, value and direction, replicated on real public LCA records and on other developers' models (§6.1–6.9); explicit decision contracts in three workflows with four declared mechanism tests (§6.10–6.11); declared transfer tests on held-out rules and independently designed layouts against strong parsers, the second powered for inference over independent record sets (§6.12–6.13); a declared test on real public records in their publishers' formats against a strengthened code-generation arm (§6.14); evidence that a deterministic compliance engine acquired the same shortcut (Appendix H); and a release in which every number in the paper is recomputed from retained replies by a script.

2 Related work

Legal NLP and compliance. Legal benchmarks (Guha et al., 2023; Chalkidis et al., 2022; Zheng et al., 2021; Hendrycks et al., 2021; Fei et al., 2024) evaluate reasoning over legal text, rule application (Zhou et al., 2025; Dou et al., 2026) and unsupported statements about the law (Dahl et al., 2024; Magesh et al., 2025). Immigration datasets are petitioner- and text-side (Shportun, 2026; Afsharrad and Mousavi, 2026). Compliance checking predates language models (Zhang and El-Gohary, 2016; Amaral et al., 2022); recent LLM work makes the model the regulated subject (Guldimann et al., 2024) or the assessor of another party's documentation (Marino et al., 2025; Yang et al., 2026; Wang et al., 2025; Sarkar, 2026; Luo et al., 2026). Absent support is already a correct answer over legal text (ContractNLI's Not-Mentioned, Koreeda and Manning, 2021; unanswerable privacy questions, Ravichander et al., 2019); what is new here is a correct answer for a missing *primary record* when obligations are assessed from a third party's operational exports.

Overclaiming under insufficient evidence. DEMM-Bench labels by construction whether agent records suffice and reports overclaims, including a "container fallacy" in which a reassuring artefact stands in for evidence (Solozobov, 2026); Partial Evidence Bench scores unsafe completeness claims, including in compliance audits, when evidence lies outside an agent's permissions (Tallam, 2026). We do not claim the directional, construction-labelled framing as new; we differ in subject, in the source of insufficiency (a missing export rather than permissions), and in manipulating proxy columns and verdict menus directly across models.

Insufficiency, distractors and answer menus. Selective prediction treats abstention as a cost (Chow, 1970; El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017); language-model abstention is studied mostly on questions (Rajpurkar et al., 2018; Yin et al., 2023; Kirichenko et al., 2025; Wen et al., 2025; Kadavath et al., 2022), and models answer rather than abstain when context is insufficient (Joren et al., 2024). NotEnoughInfo or unknown labels exist for text and tables (Thorne et al., 2018; Aly et al., 2021; Wang et al., 2021; Theologitis et al., 2026) and unanswerable database questions (Lee et al., 2022; Wang et al., 2023; Lee

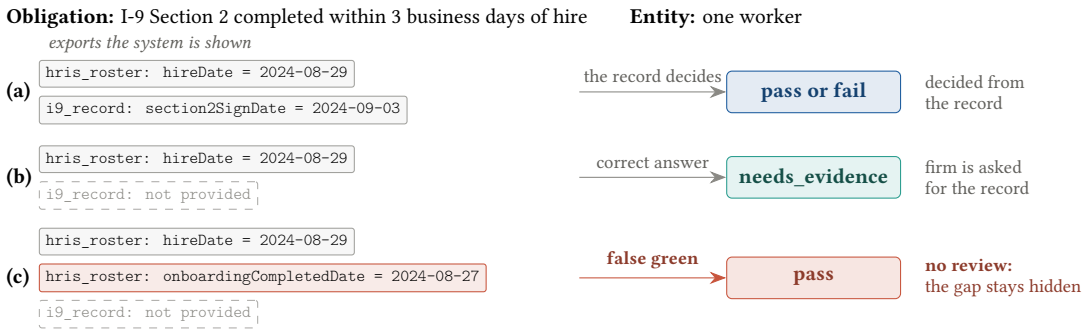


Figure 1. The task and the error. Each panel shows the exports a system is given for one obligation and one worker (values illustrative). With the I-9 record present the record decides (a); with it missing the correct verdict is needs_evidence (b); with it missing but an onboarding date beside it, a model may answer pass (c). A wrong fail sends the item to review; a wrong pass removes it from review.

et al., 2024). Irrelevant content changes answers whose label it leaves fixed (Shi et al., 2023), including irrelevant tax forms in rule application (Zhou et al., 2025), an instance of shortcut reliance (Geirhos et al., 2020); our proxy columns are the case where the distractor resembles the missing evidence and the error has a direction. Answer menus change behaviour: expanding a binary label set with "insufficient evidence" makes models more conservative (Fu et al., 2026); the presence and framing of an abstention option change how often models commit wrongly (Guo et al., 2026; Madhusudhan et al., 2025); output formats and forced choice alter answers (Tam et al., 2024; Röttger et al., 2024; Sclar et al., 2024); and grading that gives no credit for uncertainty rewards guessing (Kalai et al., 2025). In CLEAR, adding "I don't know" to multiple-choice medical questions *increased* wrong answers (Guo et al., 2026); here the abstaining verdicts are defined by whether the required record is present, and offering them lowers false greens (§6), so the effect of an abstention option depends on its definition. Our schema conditions place these effects in a compliance interface and vary the menu's parts one at a time. Pairing a language model with a symbolic solver or executor is established for logical reasoning and tables (Pan et al., 2023; Cheng et al., 2023); our remedy applies that design to evidence-bound decisions.

3 Task: decisions under an evidence contract

An instance is a set of records (CSV tables) about entities, a set of rules, and the (rule, entity) **pairs** to decide. Each rule is a **decision contract**: a predicate over named facts, together with the records that establish each fact and how to treat missing, dated and conflicting values. The correct verdict for a pair is fixed by the complete situations compatible with the visible records (Table 1). It is `pass` if every compatible completion satisfies the predicate, `fail` if every one violates it, and `needs_evidence` if both occur or the establishing records conflict with no stated priority. A missing record therefore calls for abstention only when its value could change the answer: a violation that the visible records already establish is a `fail`, whatever is missing. Labels are correct relative to the stated contract, and we make no claim that a contract reproduces professional practice (§9).

Two ways of stating a contract. In Studies 3–8 the evidence is *described*: each obligation lists the information it needs, and the model matches columns by meaning, as an employer-records reviewer would. The workforce contracts follow what the regulations make inspectable. An I-9 inspection is "an inspection of Forms I-9" on three business days' notice (8 C.F.R. 274a.2(b)(2)(ii)), and ICE's guidance makes "failure to present the Form I-9 for inspection upon request" a substantive violation (U.S. Immigration and Customs Enforcement, 2026). The LCA wage "must be paid to the employee, cash in hand, free and clear, when due" (20 C.F.R. 655.731(c)(1)), with documentation kept to prove it (655.731(b)(1)). So an HR status, an onboarding date or a salary of record does not establish these facts. In Studies 9 and 10 the evidence is *bound*: each fact names the only columns that establish it, in three workflows (workforce, purchasing and engineering acceptance). Study 10 compares the two forms on identical records.

Metrics. The **false-green rate** (FGR) is the share of unassessable pairs answered `pass`. Answering `fail` on such a pair is also wrong, but it triggers review, so it is not counted. A pair with no parsed answer can never be a false green, so every table reports coverage. FGR alone is gameable: `always_fail` scores 0 and catches every violation. Violation recall and the false-alarm rate on conforming controls (any verdict but `pass`) must therefore be reported with it (Appendix D gives trivial baselines). `unmonitored` on an unassessable pair is wrong but not a false green; in practice it acts as a second abstention (Appendix Figure 13).

4 The instrument

Rulebook. 194 obligations across ten US employment-immigration regimes, each with its check, threshold, citation and required information, built from the registry of a commercial compliance engine developed by the authors' company, whose citations its maintainers had checked against primary sources. The rulebook defines the task's scope; the experiments test four of its rules.

Generator. Labels precede data: for each pair the gener-

Table 1. The four verdicts.

Verdict	Meaning
`pass`	Every completion compatible with the records satisfies the rule
`fail`	Every compatible completion violates it
`needs_evidence`	Some compatible completions satisfy it and some violate it, or the establishing records conflict
`unmonitored`	The rule does not apply to the entity

ator draws a disposition (a planted violation, a conforming control with a share placed exactly on the legal boundary, or withheld evidence) and then writes records that decide it. It renders four rule families completely: I-9 Section 1 and Section 2 timeliness, reverification before authorisation expires, and the LCA required wage. Evidence is withheld a whole export at a time, since a blank date for a past deadline would read as "not done", a violation. A firm has twelve workers and 42 pairs. No generated pair has `unmonitored` as its correct verdict, so that verdict is exercised only on the anchor firm.

Proxy columns. For Study 3 the generator can add five roster columns that sit beside a rule's evidence without being it: an I-9 status of "Complete", an onboarding date, the current authorisation's end date, a salary of record set at the LCA offered wage, and a pay-band midpoint. They appear whether or not the export they imitate is withheld, and come from a separate random stream, so a plain firm and its proxy twin have identical labels and other exports (Appendix Figure 12). The wage rule has a native proxy even in plain firms: the LCA's offered wage beside a missing actual-wage table.

Certified labels. Every label is checked by a program written separately from the generator. For determinate labels, a reference checker written from the contracts without access to the generator reproduces all 1,176 distinct generated labels of Studies 3–8, after exposing one generator bug: a year-end observed holiday that touched no study firm. For each `needs_evidence` label we publish a **certificate**: two completions of the visible records with opposite outcomes. Each completion is written out as records and judged by the independent checker. All 3,330 such labels in Studies 3, 3c, 4 and 8 are certified, as are all 227 that turn on missing values in Study 9 (454 completions checked). Study 9's 36 conflict labels, which no completion can certify, and all its 1,008 labels agree with a second, separately written checker (§6.10). Agreement on generated packets does not show that two implementations agree everywhere: a review found an input the packets never contain, two disagreeing rows in a prioritised source, on which the generator's resolution depended on row order. Priority now ranks sources, not rows (no label changed), and on 3,000 random packets outside the generator's patterns (23,994 labels with duplicate rows, blank cells, decimal amounts and every source combination) the generator and the checker agree on every label, and shuffling rows changes none (`scripts/differential_contracts.py`). A second review found that amounts were compared as binary floating point and that candidate values stepped by one, which misjudges decimals closer than one apart (0.12 against 1.2×0.10). Amounts are now exact rationals; on the random packets with decimals, the earlier code disagreed with the checkers on 121 Study 9 labels and 3 Study 13 labels, and the corrected code on none. No published label involves a decimal, and none changed. Boundary choices are stated with their sensitivity (Appendix G). An expert workbook and two AI pilot runs of it are disclosed in Appendix I; they are not part of the validation argument.

Anchor firm. A hand-authored employer of twelve workers with eleven exports and 221 pairs, whose labels were the engine's acceptance gate, where `fail` means "requires action". It is the weakest component. Two blind AI audits of its 120 evaluated evidence requests overturned 2 and 5 labels with none in common (Cohen's $\kappa = -0.024$), and the second auditor found only 7 of its 20 violation labels shown by the exports. We use it only as a realism probe for false greens, which do not depend on violation labels (Appendix D).

5 Protocol

Each request carries the relevant exports verbatim, the obligations with their required information, and twelve pairs; the reply gives a verdict and the evidence field relied on per pair. The prompt is frozen (version 1.1.0, Appendix B); only the verdict menu changes across four conditions (Table 2).

The step from *offered* to *closed* is the **abstention effect**; from *closed* to *dashboard*, the **definition-and-warning effect**, whose two changes Study 4 separates (§6.4). Thirteen models from eight developers were run with temperature 0 and a 32,000-token cap (Appendix Table 15): six free-tier models, and seven on Google's Vertex AI paid from cloud credits. Unparseable lines are never coerced to a verdict (we report the unparseable rate), delivery failures are re-requested, and the commercial engine is not a baseline, since it and the anchor labels were developed together (Appendix C).

6 Experiments: diagnosis, then remedy

The studies fall into two parts. §6.1–6.9 diagnose the error in one domain: a preregistered study of proxy columns and output schemas (Study 3) and declared follow-ups that take the effects apart, replicate them on fresh firms, real LCA records and other developers' models. §6.10–6.12 test a remedy: explicit decision contracts in three workflows (Study 9), declared tests of why proxies persuade (Studies 10–12), and

Table 2. The schema conditions.

Condition	Verdicts	Change from the row above
offered	all four	—
removed-only	no `needs_evidence`	Abstention verdict and three sentences about it removed
closed	`pass`, `fail`	`unmonitored` removed; every pair must be decided
dashboard	`pass`, `fail`	Both verdicts redefined as "the obligation is satisfied / violated"; warning sentence removed

two declared transfer tests of the remedy on held-out rules and layouts against strong parsers (Studies 13 and 14, the second powered), and a test on real public records in their native formats (Study 15). The second part carries the paper's main claim; the first establishes the problem it addresses.

6.1 Design

Study 3 was declared in a file committed before any call (commit `e1baaeb`), with its analysis script: twelve fresh generated firms (504 pairs, 180 unassessable), each rendered plain and with proxy columns, under the four schemas, with seven models from five developers, once per cell; every cell answered at least 456 of 504 pairs. Contrasts are paired on pairs answered in both cells and averaged over models with equal weight. The declared inference is a bootstrap over the 18 (firm, rule) clusters of unassessable pairs. Four hypotheses predicted increases: **P1** the abstention effect and **P2** the definition-and-warning effect, on proxy-column firms; **P3** and **P4** the proxy-column effect under the dashboard and offered schemas.

6.2 Results

All four were confirmed (Table 3, Figure 2). The bootstrap p-values all sit at its floor (0.002 after correction), so we add an exact sign-flip permutation test, which gives Holm-adjusted $p \le 0.0013$ for all four. Two features of the design temper the headline numbers. The 18 clusters are unbalanced: the I-9 pairs (99) come from only three firms that withhold the I-9 export, and the wage pairs (81) from nine. Weighting the twelve firms equally instead of the pairs, every effect stays positive and significant (exact p from < 0.001 to 0.028), but P3 falls to +0.244, because the large I-9 effect then counts for three firms rather than 99 pairs. Rules within a firm share its records and its withholding draw, so the cluster test treats them as exchangeable where the firm test does not; we read the firm-weighted values as the conservative magnitudes and the pooled ones as specific to this mix of firms. Dropping any one model or any one firm leaves every effect positive.

False greens are not the only wrong answer on an unassessable pair. Averaged over the seven models, on proxy-column firms with abstention offered, 0.517 of unassessable pairs were answered `needs_evidence`, 0.395 `pass` and 0.045 `fail`; under the dashboard schema, where no correct answer exists, 0.713 were `pass` and 0.237 `fail` (Appendix Figure 13).

The models differ (Figure 3). The proxy-column effect under the dashboard schema is significant for all seven, and under the offered schema for five (not Gemini 3.1 Pro or Kimi K2). The abstention effect is significant for five (not Gemma 4 or gpt-oss-120b), the definition-and-warning effect for one (gpt-oss-120b). Removing only `needs_evidence` raised false greens by 0.080 pooled ($p = 0.004$). The declared interaction, a larger definition-and-warning effect on proxy-column firms, was not found (+0.048 [−0.01, 0.10], $p = 0.10$). On plain firms both schema effects are present but smaller (exploratory: +0.093 and +0.035).

6.3 Two declared follow-ups

Study 3b: a reworded prompt. Each condition had been tested in one wording. We reworded every instruction sentence of the offered and dashboard prompts, keeping meaning and verdict tokens, and re-ran the Study 3 firms with the three Google models (declared before any call; the four other developers' models were not re-run, because their cloud billing had not yet posted). All three declared tests replicated (Holm-adjusted $p = 0.0015$ each, the bootstrap's floor). Proxy columns raised the false-green rate by 0.483 under the reworded dashboard schema and by 0.301 under the reworded offered schema, against 0.440 and 0.265 for the same three models in the original wording; moving from the offered to the dashboard schema on proxy-column firms raised it by 0.340, against 0.321.

Study 3c: more I-9 firms, one column at a time. To test the I-9 effect on more than three firms, and to ask which column carries it, we drew ten fresh firms that all withhold the I-9 export and rendered each plain, with all five proxy columns, and with each I-9 proxy column alone, with the same three models (declared before any call; firm-level exact permutation tests; Figure 4). In Study 3 the I-9 effect rested on three firms: models called 0.000 to 0.023 of plain-firm I-9 pairs compliant across the four schemas, and 0.193 under the offered and 0.686 under the dashboard schema once proxy columns were present (the wage rule, whose plain firms already show the LCA's offered wage, drew false greens on 0.442 of pairs even there with abstention offered). On the ten new firms all five declared tests were confirmed (exact $p \le 0.004$, Holm-adjusted ≤ 0.010, positive in at least nine firms). All five columns together raised the I-9 false-green rate by 0.822 under the dashboard schema and by 0.302 under the offered schema. One column at a time, the onboarding date raised it by 0.509 on Section 1 and 2 pairs and the current authorisation's end date by 0.759 on reverification pairs, while the I-9 status flag raised it by only 0.097, almost

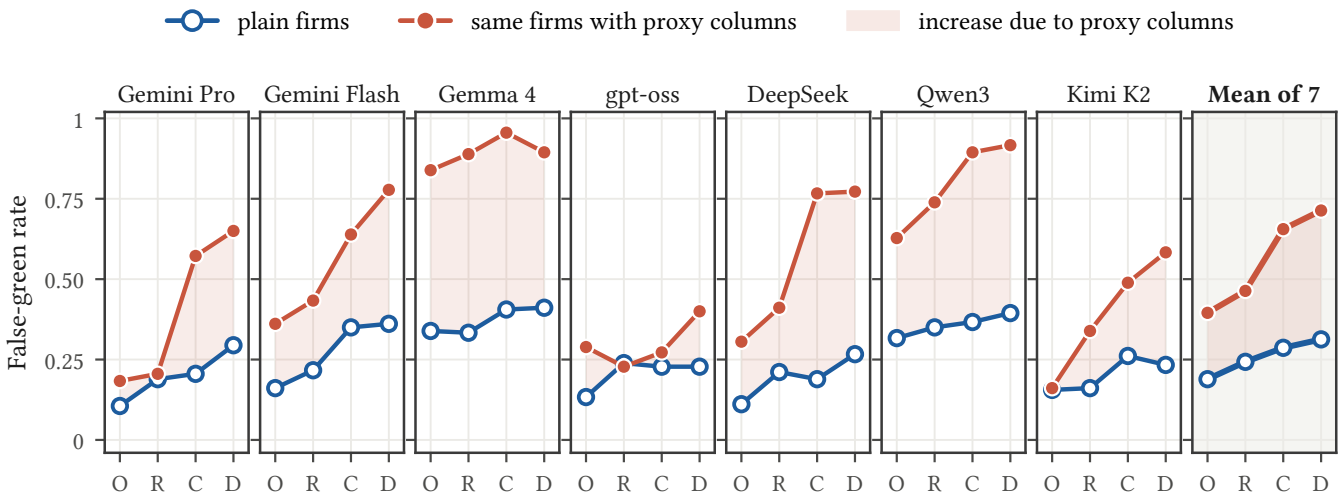


Figure 2. False-green rate on the 180 unassessable pairs of the Study 3 firms under the four schemas (O offered, R removed-only, C closed, D dashboard), for plain firms and the same firms with proxy columns. Shading marks where proxy columns raise the rate. Rates are per cell; the tests in Table 3 are paired.

Table 3. The four declared tests of Study 3, with robustness checks.

Test	Effect [95% CI]	Exact p (Holm)	Firm-weighted effect, exact p	Drop one model	Drop one firm
P1 Abstention (proxy firms)	+0.259 [+0.20, +0.33]	<0.001	+0.203 (p < 0.001)	+0.226 to +0.306	+0.227 to +0.271
P2 Definition and warning (proxy firms)	+0.084 [+0.04, +0.12]	0.001	+0.051 (p = 0.028)	+0.067 to +0.099	+0.065 to +0.091
P3 Proxy columns, dashboard	+0.427 [+0.29, +0.56]	<0.001	+0.244 (p = 0.021)	+0.407 to +0.469	+0.353 to +0.449
P4 Proxy columns, offered	+0.217 [+0.15, +0.29]	<0.001	+0.237 (p = 0.002)	+0.170 to +0.244	+0.200 to +0.230

Effect: change in false-green rate, pooled over seven models with equal weight, paired on unassessable pairs answered in both cells; interval from the declared bootstrap over 18 firm-by-rule clusters. Exact p: sign-flip permutation over those 18 clusters (all 262,144 patterns), Holm across the four tests. Firm-weighted: each of the 12 firms weighted equally, exact sign-flip over firms (4,096 patterns). Drop one: range of the pooled effect with any one model, or any one firm, left out. All columns after the first were computed after the results.

entirely through Gemma 4. Models take a date or an amount that looks like the required one as evidence far more readily than a status flag. The reverification case is the most defensible proxy: a current authorisation implies, without showing, a timely reverification.

6.4 Study 4: what the effects are made of

Study 4, declared before any call (commit `0c874fc`), reuses the ten Study 3c firms and the three Google models to take both effects apart (Table 5, Figure 5). Contrasts are paired on pairs answered in both cells, a conditional estimand; cell rates are over answered pairs, and coverage is in Appendix C.9.

Names and values. Under the offered schema, the proxy values under neutral column names (`rosterField1–5`) raised false greens by only 0.052 over plain firms, and five irrelevant columns of the same types and positions by 0.004. The named proxy columns raised them 0.249 more than their renamed twins (H1) and 0.298 more than irrelevant columns (H2). Under the dashboard schema, where every pair must be decided, the renamed values alone added 0.668. For these models, names matter when a model may abstain, and when it must decide plausible values suffice; two other developers' models do not need the names (§6.7). Gemma 4 accounts for 82% of the named-versus-renamed difference in false-green counts, so H1 rests mostly on one model.

One direction. Proxy columns whose values suggest a breach (an I-9 status of "Incomplete", onboarding ten or more days after hire, an authorisation that has ended) moved few answers to `fail`: +0.025 (H3, Holm p = 0.047), all from Gemma 4. Gemini 3.1 Pro and 3.8 Flash answered `needs_evidence` on all 330 such pairs, although they took favourable proxies as evidence of compliance; Gemma 4 answered `pass` on 89 of them, accepting the column's presence whatever it said. The proxy effect is therefore not symmetric evidence substitution: reassuring proxies are accepted and damning ones discounted, which is the direction a false green needs.

Menu parts. In a 2×2×2 factorial of the verdict menu on proxy-column firms, removing the two abstaining verdicts raised false greens by 0.384 (H4) and removing the warning by 0.093 (H6). Replacing the evidence-anchored definitions of `pass` and `fail` with bare ones changed them by −0.039 (H5, Holm p = 0.080, not confirmed), if anything downward. The parts interact (exploratory; the declaration listed interactions as descriptive): the warning mattered only without the abstaining verdicts (+0.179, against +0.007 with them), and without them bare definitions *lowered* false

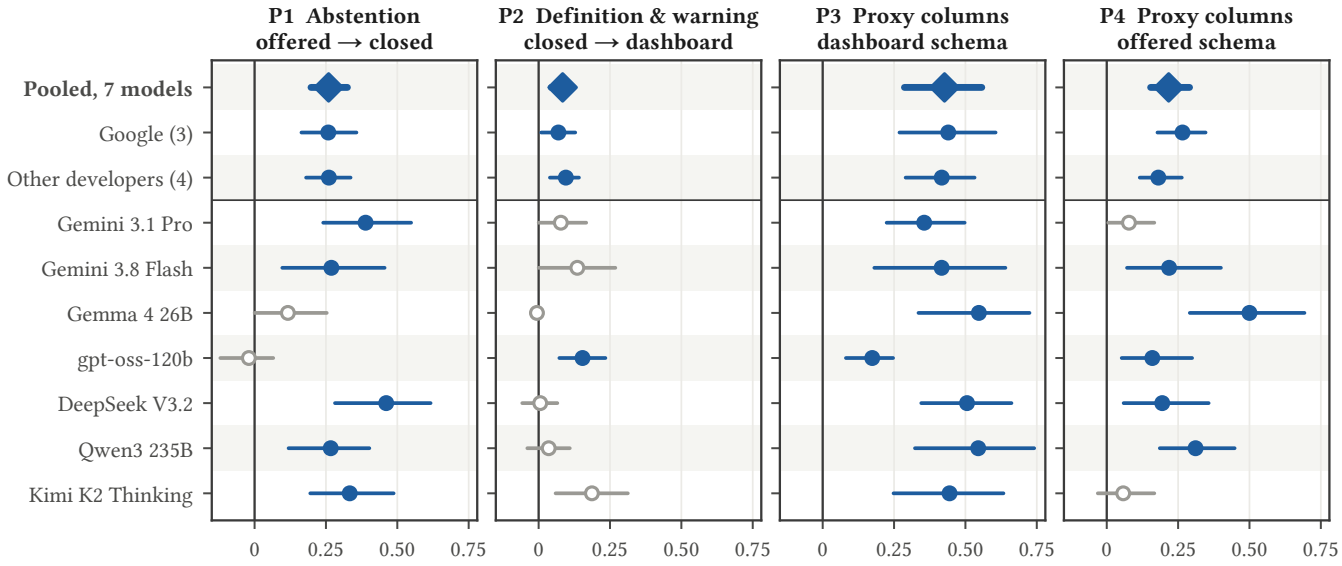


Figure 3. The declared tests per model and by developer group, with 95% bootstrap intervals. Filled markers are significant after Holm correction within their family; hollow grey markers are not. Pooled effects are robust; per-model effects vary, most of all P2.

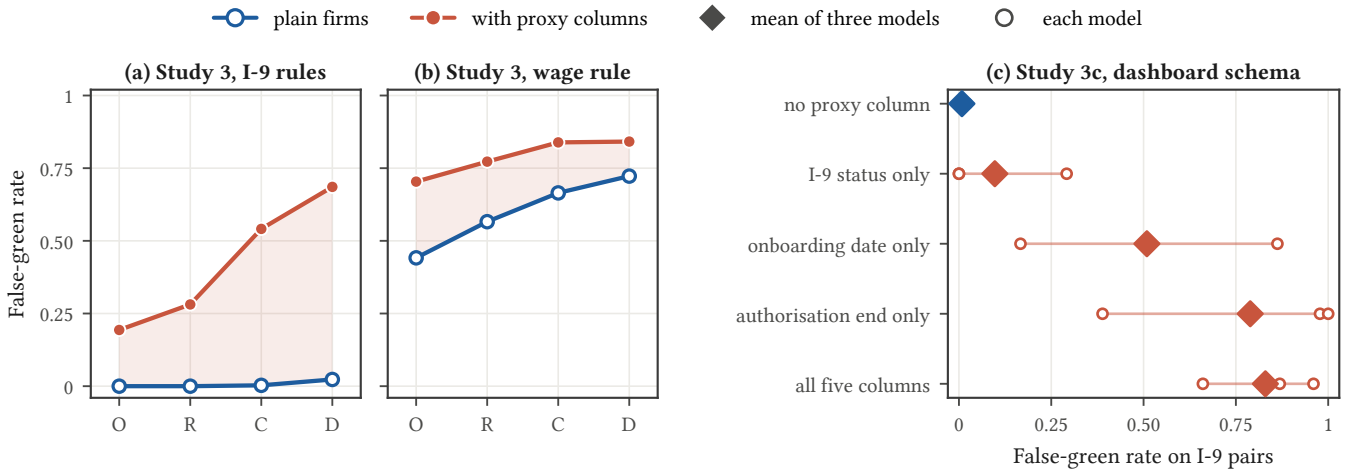


Figure 4. Where proxy columns act. (a, b) Study 3 false-green rates by rule family under the four schemas (I-9 pairs from 3 firms, wage pairs from 9). (c) Study 3c, ten further firms that withhold the I-9 export, dashboard schema: with no proxy column, with each I-9 proxy column alone (scored on the pairs it bears on) and with all five; diamonds are the mean of three models, circles the models.

greens (−0.093). Study 3's definition-and-warning effect (P2) is therefore more plausibly the warning's than the definitions', although Study 4's menus are worded differently. On plain firms no part moved false greens by more than 0.019.

An evidence-first prompt. Adding to the offered menu a requirement to name, as `<export>.<column>`, the column that is itself the required record, with two examples and a restated rule to abstain when it is absent, lowered false greens on proxy-column firms from 0.302 to 0.228 (H7, Holm p = 0.047). The change is a bundle, and the reply format still puts the verdict before the evidence field, so it does not isolate ordering. It lowered false greens for all three models: Gemini 3.1 Pro to zero, Flash from 0.115 to 0.049, and Gemma 4 from 0.770 to 0.636, the largest drop but still the highest rate; none of Gemma 4's 210 remaining `pass` answers named the I-9 export. No model's recall over all violations fell, but on plain firms Gemma 4 left 3 of 30 violations and 9 of 60 controls unanswered under it. A deterministic gate that routes a rule to `needs_evidence` when an export it needs is absent removes every false green here, but it reads the same rule-to-export mapping the generator uses to withhold evidence, so the result is guaranteed by construction: a structural baseline, not evidence for a safeguard against partial, stale or conflicting records.

Repeat runs. Running the original plain and proxy cells twice more (not the factorial, the controls or the evidence-first prompt) moved per-model false-green rates by at most 0.100, and the proxy effect stayed between 0.300 and 0.347

Table 4. The column cited as evidence on each false green, proxy-column firms.

Rule family	Schema	False greens	Citing a planted proxy column	Per model	Citing the LCA offered wage
I-9 (three rules)	offered	134	118 (88%)	50%–100%	0 (0%)
I-9 (three rules)	dashboard	462	276 (60%)	0%–100%	0 (0%)
LCA required wage	offered	364	123 (34%)	4%–77%	119 (33%)
LCA required wage	dashboard	437	124 (28%)	0%–71%	45 (10%)

Study 3 proxy-column firms, pooled over the seven models; per model: range of the share citing a planted column among models with at least five false greens. The column is the evidence field the reply line names, a self-report. The I-9 status column was never cited. Exploratory.

under the offered schema and between 0.798 and 0.822 under the dashboard schema.

6.5 Study 5: real LCA records

Study 5, declared before any call (commit `bc4a796`), replaces generated LCAs with 120 certified H-1B LCAs drawn by a fixed rule from the Department of Labor's public FY2026 Q3 disclosure file (Appendix A). A disclosure states the offered and prevailing wages but never the wages paid, so whether the required wage was paid (20 C.F.R. 655.731) cannot be decided from it; the label needs no legal interpretation. The rows form 12 firms of 10, and each worker sits on a roster whose columns follow BambooHR's export format, with or without its `PayRate` field, a salary of record set to the LCA's offered wage. A third arm adds a payroll register and an actual-wage export, making every pair decidable.

With abstention offered and no pay field, Gemini 3.1 Pro and 3.8 Flash answered `needs_evidence` on every pair they answered (Flash answered 119 of 120), while Gemma 4 called 0.825 of them compliant. Adding the roster's pay fields raised false greens by 0.236 (R1, Holm p = 0.016; paired on pairs answered in both arms, so the firm where Gemma 4 answered no salary-arm pair averages the other two models; the equally weighted all-pair rates are 0.275 and 0.469): Flash then called half the pairs compliant and cited `PayRate` on 40 of its 60, while Pro stayed at zero. Under the dashboard schema false greens were 0.861 without the pay field and rose a further 0.108 with it (R2, not confirmed); removing the abstaining verdicts raised them by 0.586 (R3, Holm p = 0.0015). Given the payroll register, Pro and Flash detected every underpaid worker with no false alarm, so their abstention was not indiscriminate; Gemma 4 detected 0.066 of them. With real LCA records inside constructed employers, the wage rule thus shows both effects of Study 3, the proxy effect with abstention offered in two of three models.

6.6 What models rely on

Every reply line names the evidence field relied on. This is the model's own report, and stated reasons need not be the cause (Turpin et al., 2023); the causal evidence is the paired contrasts of Studies 3, 3c and 4, and the citations only show what models say they used (Table 4). On proxy-column firms, 1,376 of the 2,807 false greens (49%) cite one of the five planted columns, and for the I-9 rules under the offered schema 118 of 134 (88%), almost all the onboarding date or the current authorisation's end date. On plain firms, wage-rule false greens most often cite the LCA's offered wage. Under the dashboard schema more false greens cite other fields, consistent with a schema that asks only whether the obligation is satisfied.

Detection does not bring safety. On the proxy-column firms, Gemini 3.1 Pro, Gemini 3.8 Flash, gpt-oss-120b and Kimi K2 detect at least 0.920 of violations with abstention offered, yet call 0.161 to 0.361 of unassessable pairs compliant. Across the seven models recall and false-green rate are negatively related, so higher recall helps, but none of the high-recall models comes near zero (Appendix Figure 14). Proxy columns change the false-alarm rate on conforming controls by at most 0.038 with abstention offered and 0.080 under any schema, so their cost falls mostly on the false-green side.

6.7 Study 6: two more developers

Studies 3b–5 used only Google's models. Study 6, declared before any call (commit `d03c7ee`), ran every declared Study 4 and Study 5 cell, unchanged, on DeepSeek V3.2 and GLM-5.2 (Zhipu AI), current models from other developers that Google serves on Vertex as pay-per-token models. Averaged over the two models, as declared, seven of the ten tests replicated with the same sign (Table 6):

- proxy columns raised false greens more than irrelevant columns did (+0.118);
- removing the abstaining verdicts raised them (+0.645);
- the evidence-first prompt lowered them (−0.097);
- on the real LCA records, the pay fields raised them by 0.562 with abstention offered, and removing abstention raised them by 0.892.

Averages can hide disagreement between models (Table 26 gives every model's effect):

- **Warning (H6).** The warning's effect came from DeepSeek (+0.059); for GLM-5.2 it was −0.005.
- **Names (H1).** The pooled names advantage did not replicate (−0.009) because the two models went opposite ways. Named proxies raised DeepSeek's false greens by 0.091 over renamed ones and lowered GLM-5.2's by 0.091; GLM-5.2 called 0.211 of unassessable pairs compliant from the renamed values. The pooled null shows that names are not necessary for the effect, not that names and values are equivalent.
- **Definitions (H5).** Bare definitions *raised* false greens for both models (+0.056), the reverse of the Google models'

Table 5. The seven declared tests of Study 4.

Test	Contrast	Effect [95% CI]	Firms + / −	Exact p (Holm)
H1 Names	proxy columns − same values, neutral names	+0.249 [+0.21, +0.29]	10 / 0	0.014
H2 Meaning	proxy columns − irrelevant columns	+0.298 [+0.27, +0.33]	10 / 0	0.014
H3 Direction	adverse proxies − none; outcome `fail`	+0.025 [+0.01, +0.04]	7 / 0	0.047
H4 Abstaining verdicts	removed − listed (factorial)	+0.384 [+0.37, +0.40]	10 / 0	0.014
H5 Definitions	bare − evidence-anchored (factorial)	-0.039 [-0.08, -0.00]	3 / 7	0.080
H6 Warning	absent − present (factorial)	+0.093 [+0.07, +0.12]	10 / 0	0.014
H7 Evidence-first prompt	offered − offered with evidence-first prompt	+0.074 [+0.03, +0.13]	7 / 1	0.047

Ten Study 3c firms (330 unassessable I-9 pairs), three Google models. Outcome: false green (`pass` on an unassessable pair) except H3 (`fail`). H1-H3 and H7 use the offered schema; H4-H6 average the four factorial menus on each side of the switch. All tests except H3 are on proxy-column firms. Paired, averaged within firm and model then over models; interval from a firm bootstrap; exact sign-flip over the ten firms (1,024 patterns), Holm across the seven tests. Declared before any call.

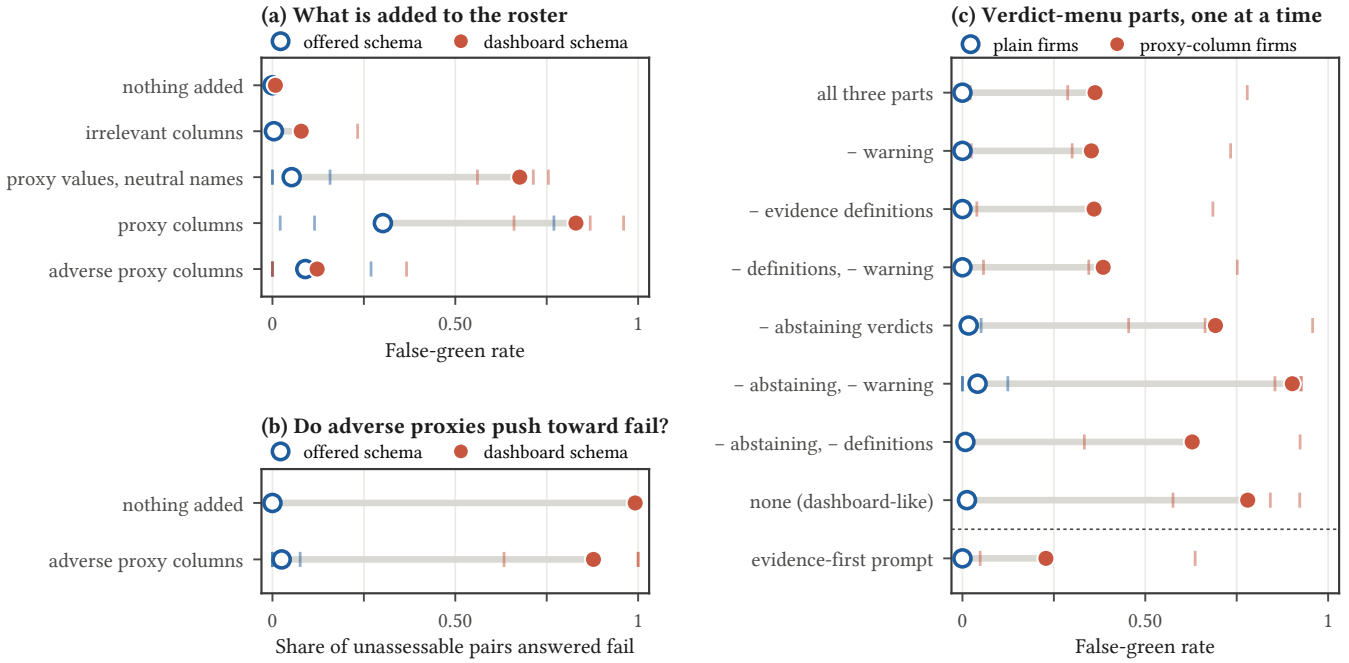


Figure 5. Study 4: ten firms that withhold the I-9 export, three models; dots are means over models, ticks individual models. (a) False-green rate by what is added to the roster, under the offered and dashboard schemas. (b) Share of unassessable pairs answered fail with nothing added and with proxies that suggest a breach. (c) The verdict menu with its three parts (abstaining verdicts, evidence-anchored definitions, warning) removed one or more at a time, and the offered menu with the evidence-first prompt, on plain and proxy-column firms.

non-significant −0.039. So the names and the wording of `pass` matter in a model-specific way, while the abstaining verdicts and the proxy columns matter for all five models.

- **Breach proxies (H3).** Both models moved some unassessable pairs to `fail` when the proxies suggested a breach (+0.064), so their substitution runs in both directions.
- **Cost of the evidence-first prompt.** For DeepSeek the prompt cut false greens from 0.121 to 0.015, but it also cut violation recall from 0.133 to 0.067 and raised false alarms on controls from 0.167 to 0.200. It traded one error for others.

Given complete records, GLM-5.2 detected every underpaid worker and DeepSeek V3.2 none.

6.8 Study 7: which pay field

Study 5's salary arm adds three roster fields at once: `PayRate`, `PayType` and `PaySchedule`. Study 7, declared before any call (commit `75a21a1`), added each part alone to the twelve real-LCA firms under the offered schema, on all five models.

- **`PayRate` alone** raised false greens by 0.227 (W1, Holm p = 0.001, positive in all twelve firms): +0.775 for DeepSeek V3.2, +0.182 for Gemma 4, +0.160 for Gemini 3.8 Flash, and zero for Gemini 3.1 Pro and GLM-5.2.
- **Pay type and schedule alone** added 0.067 (W2, not confirmed).
- **The combination.** For two models it went further than the salary alone: relative to `PayRate` alone, all three fields raised Flash's false greens by 0.342 and GLM-5.2's by 0.244.

Table 6. The declared tests of Studies 4 and 5 on two model sets.

Test	Gemini 3.1 Pro, 3.8 Flash, Gemma 4	DeepSeek V3.2, GLM-5.2	Replicated
H1 Names (proxy – renamed)	+0.249 (p 0.014)	-0.009 (p 0.828, n.s.)	no
H2 Meaning (proxy – irrelevant)	+0.298 (p 0.014)	+0.118 (p 0.014)	yes
H3 Direction (adverse, `fail`)	+0.025 (p 0.047)	+0.064 (p 0.023)	yes
H4 Abstaining verdicts removed	+0.384 (p 0.014)	+0.645 (p 0.014)	yes
H5 Bare definitions	-0.039 (p 0.080, n.s.)	+0.056 (p 0.014)	no (opposite sign)
H6 Warning removed	+0.093 (p 0.014)	+0.028 (p 0.043)	yes
H7 Evidence-first prompt	+0.074 (p 0.047)	+0.097 (p 0.014)	yes
R1 Salary of record, offered (real LCAs)	+0.236 (p 0.016)	+0.562 (p 0.002)	yes
R2 Salary of record, dashboard (real LCAs)	+0.108 (p 0.219, n.s.)	+0.046 (p 0.188, n.s.)	both n.s.
R3 Dashboard – offered (real LCAs)	+0.586 (p 0.001)	+0.892 (p 0.001)	yes

Effects on the false-green rate (H3: on `fail`), each model weighted equally; Holm-adjusted exact sign-flip p over firms (ten for H, twelve for R), Holm within each study and model set as declared. Study 6 ran the declared Study 4 and 5 cells unchanged on two more developers' models. Contrasts are paired on pairs answered in both cells; where a model answered none of a firm's pairs in a cell, that firm's contrast averages the remaining models. Per-model effects are in Table 26.

Test	Effect
F1: proxy columns over none, offered	+0.224
F2: proxy columns over irrelevant columns	+0.224
F3: dashboard over offered, on proxy firms	+0.608
F4: renamed values over none, dashboard	+0.700

The salary of record alone induces false greens in three of five models, and the surrounding pay context strengthens the response in some. Study 7's five-model estimate is not a share of Study 5's three-model one: restricted to Study 5's three models, `PayRate` alone gives +0.108 against +0.236 for the three fields together. GLM-5.2 answered 100 of 120 pairs in the `PayRate` arm.

6.9 Study 8: fresh firms, repeat runs and two wage controls

Study 6 reused the Study 3c firms and ran once; Study 5's `PayRate` equalled the LCA's offered wage; and its complete arm made the greater-of rule trivial. Study 8, declared before any call (commit `ec2a325`), addressed all three.

Fresh firms, repeat runs. Ten new firms that withhold the I-9 export were rendered plain, with proxy columns, with irrelevant columns and with renamed proxy values. All five models ran them twice under both schemas. All four declared tests were confirmed (Holm p = 0.010 each), positive in all ten firms. The pooled estimates from the two runs agreed to within 0.02, though single models moved more (GLM-5.2's F4 by 0.076):

The proxy effect under the offered schema was positive for every model, from +0.038 (Gemini 3.1 Pro) to +0.795 (Gemma 4). Proxy columns are now tested on 32 firms.

Wage controls. On the real-LCA firms:

- **G1.** `PayRate` set 7% above the LCA's offered wage still raised false greens by 0.247 (Holm p = 0.006, positive in eleven firms), so the effect does not depend on the roster repeating the LCA's number.
- **G2.** A complete arm whose actual-wage export exceeds the offered wage contains underpayments visible only under the greater-of rule. We found no significant pooled difference between recall on these 38 underpayments and on the 24 below the offered wage (G2, −0.028 [−0.115, +0.053], ten firms, not confirmed); this does not show the two are detected equally often. Gemini 3.1 Pro and GLM-5.2 detected every greater-of-only underpayment and Flash 0.947. DeepSeek V3.2 detected 0.526 of them against 0.667 of those below the offered wage, and also flagged 0.569 of correctly paid workers.

The lowest coverage in Part A was 291 of 330 pairs (GLM-5.2 again exceeding its output budget on renamed values under the dashboard schema).

6.10 Study 9: explicit decision contracts in three workflows

Studies 3–8 test one workflow, withhold whole exports, and leave a record-presence gate trivially exact. Study 9, declared before any call (commit `081290f`), states every rule as a decision contract and moves to three workflows:

- **Workforce:** a Section 2 deadline and the greater-of wage.
- **Purchasing:** spend within a signed approval, and approval before the order.
- **Engineering acceptance:** measured load at or above specification, and testing before shipment.

Each contract names the only columns that establish each fact, and every entity carries reassuring non-authoritative columns: an approval status of "Approved", a QA status of "Pass", a target load, an onboarding date. All 2,304 labels of Studies 9 and 11 agree with a second checker written independently from a plain-language specification. Study 9's 227 missing-value labels each have two completions with opposite outcomes, all 454 of them verified by that checker; its 36 conflict labels are checked by agreement (§4).

Part A: names and menus. Six packets per workflow, each rendered with familiar or neutral table and column names (identical values), under the offered or forced menu. Once the contract named the establishing columns, unsupported `pass` on missing-evidence pairs fell to at most 0.009

for every model under the offered menu, and to 0.014–0.120 under the forced menu. None of the four declared predictions held (Table 7):

- **P1:** familiar names did not raise unsupported `pass` (−0.014, not confirmed);
- **P2:** neutral names did not raise abstention on decidable pairs;
- **P3:** the forced menu did not amplify a familiarity effect;
- **P4:** neither held-out workflow showed P1.

The coverage of Gemma 4 and GLM-5.2 in Part A was 0.70 and 0.83 after retries (Appendix C.9); contrasts are paired on answered pairs, so for these two models a low error rate on answered pairs coexists with substantial incompletion.

Part B: harder packets and a remedy. Sixteen entities per packet, two of each of eight patterns:

- complete records;
- a reassuring proxy beside a blank fact;
- a missing row;
- a blank fact;
- a redundant second source;
- a violation established despite a gap;
- a conflict resolved by stated priority;
- a conflict with no resolution.

Where a workflow cannot express a pattern, the slot holds complete records: the violation-despite-a-gap slots in purchasing and engineering, and the redundant-source and resolved-conflict slots in engineering, whose facts each have a single source.

The remedy separates reading from deciding. The model reports each fact with the column it came from, and a deterministic solver accepts a value only from a column the contract names and applies the rule. Against the evidence-first prompt (Table 7, Figure 6), it:

All three declared tests were confirmed, and the remedy did better in every packet on unsupported `pass` and in 17 of 18 on correct decisions. It also beat a row-level presence gate added to the same prompt by 0.171 in correct decisions (I3). The gate lowered unsupported `pass` only to 0.060, because unresolved conflicts and redundant sources defeat a presence check.

The remedy sends more items to follow-up than the prompt (0.658 against 0.629), close to the 0.649 that correct decisions require. The prompt's lower share comes from violations hidden as `pass`.

Per model, the gains come mostly from Gemma 4 (unsupported `pass` 0.229 to 0.007) and DeepSeek V3.2 (0.108 to 0.002). Gemini 3.8 Flash's correct decisions fell from 1.000 to 0.977, and GLM-5.2's unsupported `pass` rose from 0.002 to 0.004 (Table 27). The remedy is a safeguard for models that accept proxies; for a model already near the floor it adds little and can cost correct decisions.

The decision layer depends on extraction. The solver checks that a value is attributed to a named column, not that it matches the record. We therefore audited every fact slot the rules need in the retained replies. Of 6,240 slots, 6,098 were right: the record's value from a column that holds it, `ABSENT` where the record has none, or `CONFLICT` where it conflicts. Of the errors, 101 concern facts the record establishes (47 omitted, 43 reported absent, 11 reported conflicting), 7 cite a column that does not hold the value, 3 invent a value for a missing fact, and 26 miss a conflict. Gemma 4 recognised 12 of 36 conflicts; the other models 35 or 36. Of the 126 pairs with a slot error, 94 received a wrong verdict.

A correction. The first version of the solver had a defect, found in review: when both dates of a rule were reported absent, a single fallback value made them equal and the rule passed. It produced 24 of the remedy's then 31 unsupported `pass` answers (0.011). After the fix, with a regression test, the same retained replies give 0.002. The labels were unaffected.

What this shows and what it does not. A program reading the named columns, here the independent checker, is exact on these packets, because their schema is known. The comparison therefore shows how to structure a language-model system that must decide from records, not that a language model is needed when a fixed schema allows a program. The approach follows earlier work that pairs a language model with a symbolic executor (Pan et al., 2023; Cheng et al., 2023).

6.11 Studies 10–12: why proxies persuade

Study 9 removed most false greens that Studies 3–8 found in abundance. Three declared studies (commits `fb929fc`, `efd173f`, `a0ceb9c`) tested candidate reasons on the Study 9 Part A packets, with the same five models.

- **Binding versus describing (Study 10).** The rules listed the information needed instead of naming columns, with the binding sentence removed. Unsupported `pass` rose by only 0.002 under the offered menu and 0.043 under the forced menu (B1–B3, none confirmed; the forced-menu interval, [+0.013, +0.073], excludes zero before correction). Binding does not explain the offered-menu result; a smaller role under forced choice is not ruled out.
- **Visibility of the gap (Study 11).** The deciding records were removed as blank cells, as missing rows or as an absent table. All three forms produced almost no false greens (G1–G3, at most +0.018, none confirmed). A missing table alone does not explain Studies 3–8.
- **The reading rule (Study 12).** The Studies 3–8 prompt tells the model that "column names will not match the wording of the obligations" and to "match them by meaning". *Removing* that sentence from the original prompt on the Study 3c proxy-column firms lowered false greens by 0.085 (M2, Holm p = 0.004, all ten firms): Gemma 4 from 0.770 to 0.491, and GLM-5.2 from 0.115 to 0.036. Against a re-run of the original prompt made at the same time, declared as an addendum, the estimate was +0.090 [+0.051, +0.129] (exact p = 0.006, nine of ten firms). *Adding* the sentence to the contract prompts produced no detectable

Measure	Evidence-first prompt	Extraction + solver	Program reading the named columns
Unsupported `pass` (I1)	0.068	0.002	0.000
Correct decisions on decidable pairs (I2)	0.818	0.967	1.000
Unsupported `fail`	0.034	0.006	0.000
Coverage	0.978	0.997	1.000
Unnecessary abstention	0.053	0.031	0.000
Share needing follow-up (`fail`, abstain or unanswered)	0.629	0.658	0.649

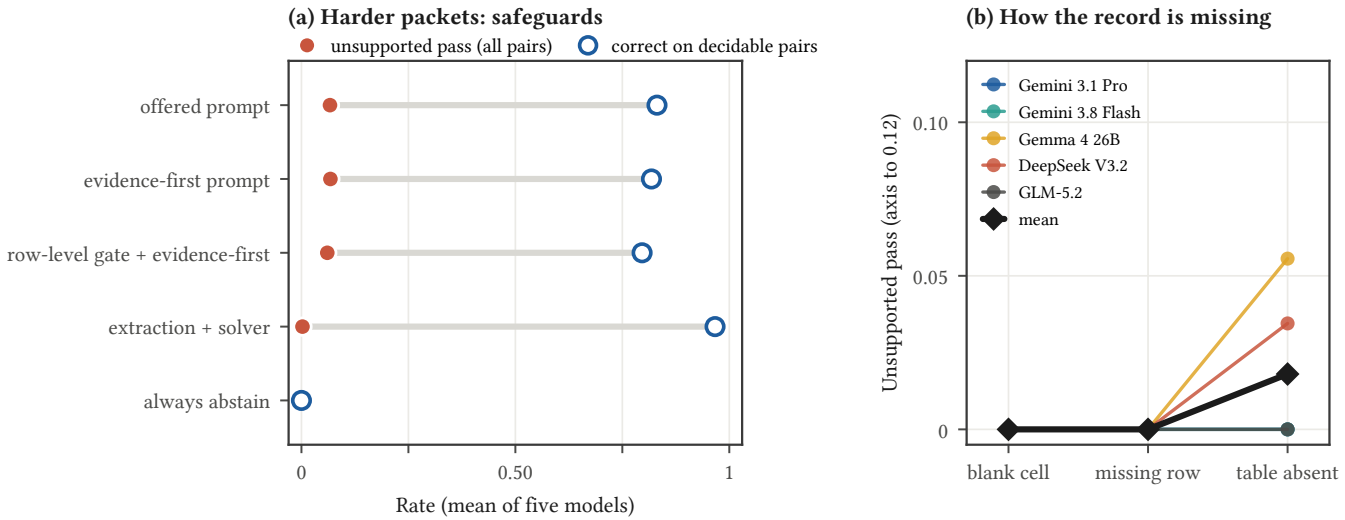


Figure 6. (a) Study 9 Part B, harder packets: unsupported pass over all pairs and correct decisions on decidable pairs, mean of five models, for each safeguard. (b) Study 11: unsupported pass on the same missing-evidence pairs when the deciding records are a blank cell, a missing row or an absent table; lines are models, the diamond the mean.

change (M1, +0.002).

Several of these tests ran near a floor of false greens, so a null result is weak evidence that a factor is irrelevant. The instruction a deployer writes to cope with mismatched column names therefore contributes to proxy acceptance, but it accounts for only part of the difference between the employer-audit framing of Studies 3–8 and the contract framing of Study 9. We do not have a complete account. What reliably reduces unsupported decisions in our data is the architectural change of Part B, not a wording.

6.12 Study 13: held-out rules and record layouts

Study 9's remedy was tested on the rules and layout it was designed with, where a program reading the named columns was exact. Study 13, declared before any call (commit `527a33d`), asks whether the remedy transfers to rules and record layouts it was not designed on, and whether reading by a language model adds anything over a strong rule-based parser when the layout is not known in advance.

Task. Six new decision rules, two per workflow, with shapes absent from Studies 9–12: a two-sided 90-day window, a posting deadline with a 10-business-day duration, a four-fact conjunction, a three-date ordering, a 365-day validity window and a 1.2× threshold; five have a prioritised fact. Packets have sixteen entities over twelve patterns, three of them new conflict structures (disagreeing duplicates in a fallback source, which priority resolves; in the deciding source, which it does not; in a single-source fact). All 480 labels agree with a checker written by a separate agent from the specification alone, all 198 certificate completions verify, and on 3,000 random packets outside the generator's patterns the two agree on all 18,000 labels.

Layouts. The same records in six layouts. L1–L3, the development layouts, are a canonical CSV, a CSV with other headers and value formats, and one line per field. L4–L6 were designed by a second separate agent from a brief and kept out of the repository until the parser below was frozen: a nested YAML export, a register of prose entries ("supplier billed USD 185.5 thousand; billed for one hundred and four (104) units"), and a fixed-width report with abbreviated headers, loads in newtons and amounts in thousands. Each test packet (two seeds per workflow) appears in all six, so layouts are paired on content.

Baselines. A third separate agent wrote a rule-based parser from the specification and development packets in L1–L3, asked to be as robust as it could to unseen layouts without a language model. It handles JSON, XML, HTML, spreadsheets, fixed-width and Markdown tables, key–value blocks and prose, with synonym matching, unit conversion and abstention, and is exact on the development packets. Its facts go to the same solver.

A defect and a replication. In the first run the rule state-

Table 7. The declared tests of Studies 9–12.

Study	Test	Effect [95% CI]	Packets + / –	Holm p	Result
9	P1 Familiar – neutral names: unsupported pass	-0.014 [-0.026, -0.004]	1 / 7	0.273	not confirmed
9	P2 Neutral – familiar names: abstention on decidable pairs	-0.018 [-0.048, +0.017]	4 / 9	0.638	not confirmed
9	P3 Forced – offered menu: familiarity effect	-0.056 [-0.119, +0.004]	4 / 11	0.362	not confirmed
9	P4 P1 on a held-out workflow (purchasing)	-0.011 [-0.022, +0.000]	0 / 2	0.638	not confirmed
9	P4 P1 on a held-out workflow (engineering)	-0.036 [-0.053, -0.019]	0 / 5	0.273	not confirmed
9	I1 Extraction + solver – evidence-first: unsupported pass	-0.065 [-0.087, -0.049]	0 / 18	<0.001	confirmed
9	I2 Extraction + solver – evidence-first: correct decisions	+0.149 [+0.113, +0.188]	17 / 0	<0.001	confirmed
9	I3 Extraction + solver – row gate + evidence-first: correct decisions	+0.171 [+0.126, +0.215]	17 / 0	<0.001	confirmed
10	B1 Described – bound evidence, offered	+0.002 [+0.000, +0.006]	1 / 0	1.000	not confirmed
10	B2 Described – bound evidence, forced	+0.043 [+0.013, +0.073]	10 / 2	0.067	not confirmed
10	B3 Described – bound, held-out workflows	+0.003 [+0.000, +0.008]	1 / 0	1.000	not confirmed
11	G1 Table absent – row missing	+0.018 [+0.000, +0.044]	2 / 0	1.000	not confirmed
11	G2 Row missing – blank cell	+0.000 [+0.000, +0.000]	0 / 0	1.000	not confirmed
11	G3 Table absent – blank cell	+0.018 [+0.000, +0.047]	2 / 0	1.000	not confirmed
12	M1 Reading rule added to contract prompts	+0.002 [+0.000, +0.006]	1 / 0	1.000	not confirmed
12	M2 Reading rule kept – removed, original prompt	+0.085 [+0.056, +0.113]	10 / 0	0.004	confirmed

Five models (Gemini 3.1 Pro, Gemini 3.8 Flash, Gemma 4 26B, DeepSeek V3.2, GLM-5.2). Unit: the packet (Study 12 M2: the firm); paired contrasts averaged over models with equal weight; exact sign-flip test (Study 9: Monte Carlo above 20 packets), packet bootstrap interval, Holm within each study. All declared before any call of that study.

ment printed some fact names with their unit ("measured load (kN)"); Gemini 3.1 Pro, Gemini 3.8 Flash and GLM-5.2 copied the unit into the name, the declared scorer matches names exactly, and the solver treated those facts as absent. Under the declared scoring, H1 and the non-inferiority test N1 were confirmed and H2 and H3 were not (+0.093 and +0.037); matching names after removing the unit, post hoc, gave +0.289 and +0.232. We corrected only the statement and declared a replication before any of its calls (commit `162faa4`); it is the run reported below, and both runs are in Table 9.

Results (held-out layouts, replication). Separating reading from deciding transferred (Tables 8 and 9, Figure 7):

- **H1, safety.** Against the evidence-first prompt, unsupported `pass` fell by 0.111 [-0.125, -0.097], in all 18 packets (Holm $p < 0.001$).
- **H2, value of learned reading.** Against the rule-based parser, correct decisions rose by 0.295 [+0.172, +0.419] (Holm $p < 0.001$), and for every model the per-model interval excludes zero (Table 28). The parser was exact on the development layouts and fell to 0.818, 0.794 and 0.418 on L4, L5 and L6; extraction stayed between 0.958 and 0.980.
- **H3, correct decisions.** Against the prompt, correct decisions rose by 0.230 [+0.212, +0.246], in all 18 packets.
- **N1, safety against the parser.** The parser is safer: unsupported `pass` 0.002 against 0.012. The difference, +0.010 [+0.006, +0.014], lies inside the declared margin of +0.02.

These tests overstate the independent evidence. The eighteen held-out inputs are six contents, each rendered in three layouts, and the declared tests treated renderings as independent. Clustered by content, all three effects point the predicted way in all six contents, but with six units the smallest attainable exact p is 0.031, and after Holm correction $p = 0.094$ for each: the replication is suggestive, not confirmed, at the level of contents. The pooled safety criterion still holds clustered by content, but not per model: Gemma 4 and DeepSeek V3.2 made more unsupported `pass` decisions than the parser. Study 14 was designed to settle these points.

Where each method fails. On the development layouts, where it was written, the parser beats extraction by 0.050 in correct decisions; a known schema still favours a program. The models' residual errors are concentrated in conflicts: extraction decided 0.733 of pairs correctly where the deciding source holds disagreeing duplicates and 0.756 where a single-source fact does, against 1.000 and 1.000 for the parser. In a slot audit, 7,452 of 7,680 extracted facts were right. Using the parser where it reports every fact and extraction elsewhere, an exploratory combination, gave 0.987 correct decisions with unsupported `pass` of 0.013.

By model. As in Study 9, the gains over the prompt come from the two models that accept proxies. Gemma 4's unsupported `pass` fell from 0.392 to 0.043 and DeepSeek V3.2's from 0.222 to 0.016; Gemini 3.1 Pro, Gemini 3.8 Flash and GLM-5.2 made no unsupported `pass` under either and reached 1.000 correct decisions with extraction. Extraction used fewer output tokens than the prompt for the three reasoning models (Gemini 3.8 Flash 342 against 1114 per pair) and about twice as many for the two that answer tersely.

6.13 Study 14: a powered transfer test

Study 13's inference rested on six contents and its layouts on one designer. Study 14, declared before any call (commit `58adb04`), tests the same claims on 30 fresh contents, ten per workflow, sized by a power analysis from Study 13: with its effects halved and modelled as normal, power was at least 0.94 for H1–H3; H4 had no Study 13 counterpart and was not powered. The rules, solver and prompts are Study 13's, frozen.

Table 8. Study 13 replication: every system on the held-out layouts.

System	Unsupported pass	Unsupported fail	Correct decisions	Correct, development layouts	Unnecessary abstention	Follow-up share
Evidence-first prompt	0.123	0.021	0.742	0.743	0.084	0.605
Extraction + solver	0.012	0.008	0.971	0.950	0.028	0.684
Rule-based parser + solver	0.002	0.016	0.677	1.000	0.307	0.793
Parser, else extraction (exploratory)	0.013	0.010	0.987	0.994	0.011	0.678
Program reading the named columns	0.000	0.000	0.000	0.333	1.000	1.000
Always abstain	0.000	0.000	0.000	0.000	1.000	1.000

Replication run; held-out layouts L4–L6 except the development column (L1–L3). Model systems: mean of five models. Unsupported pass and fail: over all pairs; correct decisions and unnecessary abstention: over decidable pairs; follow-up: share of pairs answered `fail`, abstaining or unanswered.

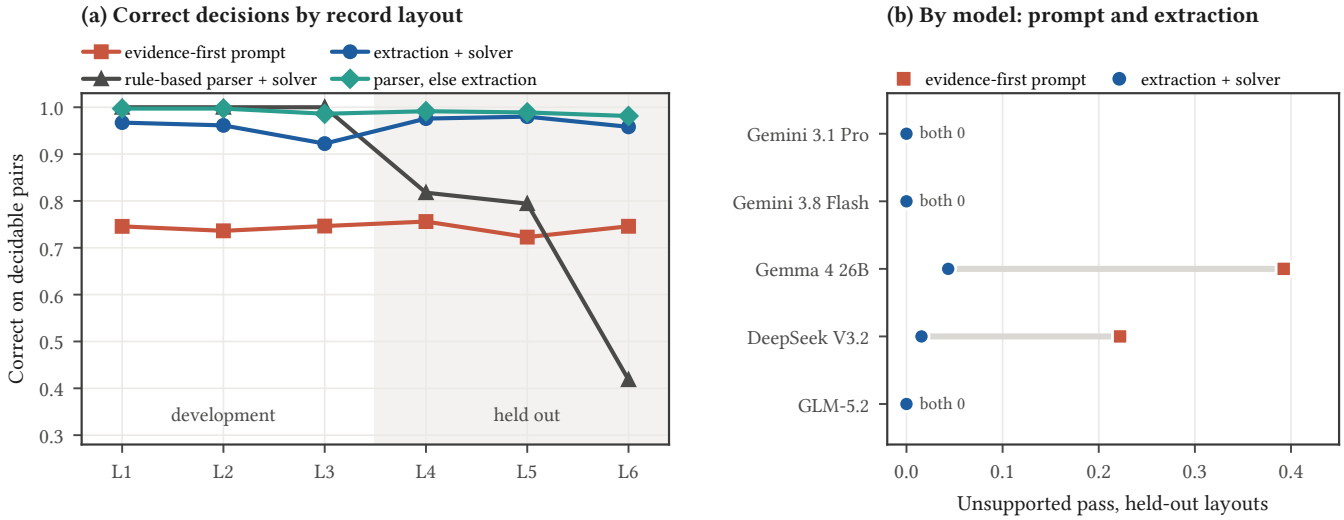


Figure 7. Study 13, replication. (a) Correct decisions on decidable pairs by record layout, mean of five models for the model-based systems; L1-L3 were available to the parser's author, L4-L6 were held out from it. (b) Unsupported pass on the held-out layouts for each model under the evidence-first prompt and extraction + solver.

Layouts and baselines. Six new layouts came from three designers who worked separately from one another and from us, each with a different direction: public and standard formats (a pipe-delimited disclosure-style export with many unrelated columns; an X12-style interchange), human documents (HTML form sheets; an e-mail export with decoy header dates and some relative dates) and system exports (an SQL dump with epoch days, cents and decanewtons; a pivoted table with ISO week dates). Each content appears in two layouts from different designers, so the unit of inference is the content. Study 13's parser author improved the parser with all six Study 13 layouts as development data, and it was frozen before the new layouts entered the repository; that author also practised on further formats of their own. A second learned baseline asks the model to write a parser for the packet in front of it; the program runs sandboxed and its facts go to the same solver. All 960 labels agree with the independent checker.

Results. All four declared tests reached Holm $p < 0.001$ over contents (Table 11, Figure 8):

- **H1, safety.** Against the evidence-first prompt, extraction lowered unsupported `pass` by 0.094 [-0.102, -0.085], in all 30 contents.
- **H2, value of learned reading.** Against the rule-based parser, extraction made 0.544 [+0.420, +0.667] more correct decisions. The parser, exact on all six layouts it was developed on, reached 0.349 on the new ones, and the improved parser did worse than its predecessor (0.395).
- **H3, the hybrid.** Using the parser where it reads every fact and extraction elsewhere beat the parser by 0.565 [+0.448, +0.670], reaching 0.914.
- **H4, a model-written parser.** Extraction made 0.172 [+0.127, +0.219] more correct decisions than code generation. Of 300 replies, 290 contained a program and 268 ran without error. H4 compares extraction with this configuration only: one program per packet under a 32,000-token cap (ten GLM-5.2 replies reached it), with no repair after execution. A program reused on another packet in the same layout decided 0.341 of decidable pairs correctly, but that figure mixes reading with an interface defect: the prompt listed the pairs and no list was supplied at run time, so 160 of the programs answered the original packet's pairs (102 had copied its identifiers) and only 102 answered the new packet's.

By layout. Extraction beat the parser in five of six layouts (exact $p = 0.062$ over layouts, a check rather than a test). The exception was B1, the HTML form sheets, which the parser read perfectly. Extraction ranged from 0.750 to 0.956 across

Table 9. The declared tests of Study 13.

Run	Test	Effect [95% CI]	Packets + / –	Holm p	Result
First run	H1 Extraction – evidence-first: unsupported pass	-0.114 [-0.125, -0.102]	0 / 18	<0.001	confirmed
First run	H2 Extraction – rule-based parser: correct decisions	+0.093 [-0.070, +0.247]	12 / 6	0.541	not confirmed
First run	H3 Extraction – evidence-first: correct decisions	+0.037 [-0.027, +0.101]	11 / 7	0.541	not confirmed
First run	N1 Extraction – parser: unsupported pass, bound < +0.02	+0.010 [+0.005, +0.013]	15 / 1	–	confirmed
First run, post hoc names	H1 Extraction – evidence-first: unsupported pass	-0.113 [-0.124, -0.101]	0 / 18	<0.001	post hoc
First run, post hoc names	H2 Extraction – rule-based parser: correct decisions	+0.289 [+0.159, +0.417]	12 / 6	<0.001	post hoc
First run, post hoc names	H3 Extraction – evidence-first: correct decisions	+0.232 [+0.213, +0.252]	18 / 0	<0.001	post hoc
First run, post hoc names	N1 Extraction – parser: unsupported pass, bound < +0.02	+0.011 [+0.007, +0.015]	15 / 1	–	post hoc
Replication	H1 Extraction – evidence-first: unsupported pass	-0.111 [-0.125, -0.097]	0 / 18	<0.001	confirmed
Replication	H2 Extraction – rule-based parser: correct decisions	+0.295 [+0.172, +0.419]	12 / 6	<0.001	confirmed
Replication	H3 Extraction – evidence-first: correct decisions	+0.230 [+0.212, +0.246]	18 / 0	<0.001	confirmed
Replication	N1 Extraction – parser: unsupported pass, bound < +0.02	+0.010 [+0.006, +0.014]	17 / 1	–	confirmed
Replication, by content (post hoc)	H1 Extraction – evidence-first: unsupported pass	-0.111 [-0.129, -0.094]	0 / 6	0.094	not confirmed
Replication, by content (post hoc)	H2 Extraction – rule-based parser: correct decisions	+0.295 [+0.210, +0.380]	6 / 0	0.094	not confirmed
Replication, by content (post hoc)	H3 Extraction – evidence-first: correct decisions	+0.230 [+0.211, +0.248]	6 / 0	0.094	not confirmed

Held-out layouts L4–L6, eighteen packets; five models averaged with equal weight; exact sign-flip test, packet bootstrap interval, Holm over H1–H3. N1 is declared non-inferiority (upper bound below +0.02). The first run's rule statement printed some fact names with their unit; the post-hoc rows match names after removing it, and the replication corrected the statement before any call (Section 6.12). The last rows treat each of the six contents as one unit (its three layouts averaged), a correction made after review.

Table 10. Study 14: every system, all sixty renderings.

System	Unsupported pass	Unsupported fail	Correct decisions	Coverage	Unnecessary abstention	Follow-up share
Evidence-first prompt	0.106	0.020	0.717	0.988	0.145	0.625
Extraction + solver	0.013	0.018	0.893	0.997	0.089	0.695
Hybrid: parser, else extraction	0.015	0.019	0.914	0.999	0.068	0.684
Code generation: model-written parser + solver	0.004	0.007	0.721	0.850	0.126	0.759
Rule-based parser + solver	0.003	0.005	0.349	1.000	0.644	0.886
Study 13 parser + solver	0.000	0.001	0.395	1.000	0.605	0.874
Always abstain	0.000	0.000	0.000	1.000	1.000	1.000

All sixty renderings; model systems are the mean of five models. Measures as in Table 8.

layouts and was weakest on the SQL dump with epoch dates.

Safety, per model (declared criterion). The parser made almost no unsupported `pass` decisions (0.003). Extraction met the declared margin (an excess below 0.01) for Gemini 3.1 Pro, Gemini 3.8 Flash and GLM-5.2, and failed it for Gemma 4 (excess +0.045 [+0.033, +0.057]) and DeepSeek V3.2 (+0.012 [+0.006, +0.017]); the hybrid did the same (Table 29). For the three models that met it, extraction decided between 0.994 and 0.999 of decidable pairs correctly; for Gemma 4 and DeepSeek V3.2, 0.704 and 0.769. Code generation was safer for those two models but decided fewer pairs correctly. For a model that already reads well the architecture adds little: Gemini 3.8 Flash under the prompt alone made no unsupported `pass` and decided 0.999 of decidable pairs correctly, against 0.994 with extraction.

Where errors remain. As in Study 13, conflicts: extraction decided 0.743 of pairs correctly where the deciding source holds disagreeing duplicates, against 0.983 for the parser, which abstains on what it cannot read. In a slot audit, 22,705 of 25,600 extracted facts were right, against 19,127 of 25,600 for code generation.

6.14 Study 15: real public records in their native formats

Studies 13 and 14 render generated records. Study 15, declared before any call (commit `5f2d184`), applies the frozen

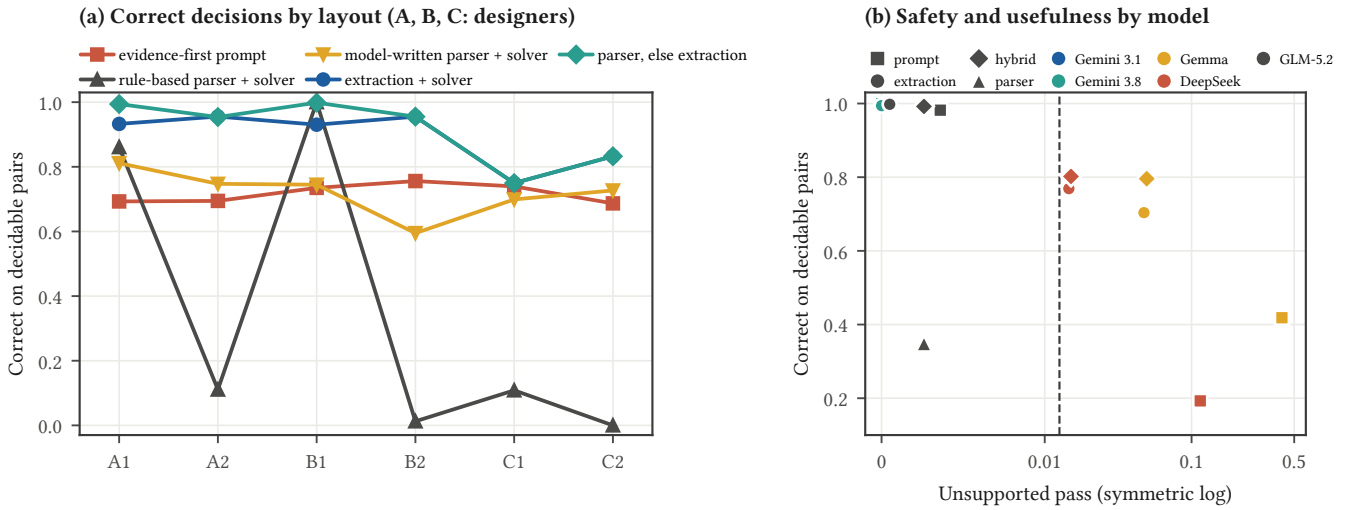


Figure 8. Study 14. (a) Correct decisions on decidable pairs by layout; letters name the three designers; model systems are the mean of five models. (b) For each model, unsupported pass against correct decisions under the prompt, extraction and the hybrid, with the rule-based parser; the dashed line marks the declared safety margin, 0.01 above the parser.

Table 11. The declared tests of Study 14.

Test	Effect [95% CI]	Contents + / −	Holm p	Result
H1 Extraction − evidence-first: unsupported pass	-0.094 [-0.102, -0.085]	0 / 30	<0.001	confirmed
H2 Extraction − parser: correct decisions	+0.544 [+0.420, +0.667]	26 / 4	<0.001	confirmed
H3 Hybrid − parser: correct decisions	+0.565 [+0.448, +0.670]	26 / 2	<0.001	confirmed
H4 Extraction − code generation: correct decisions	+0.172 [+0.127, +0.219]	26 / 4	<0.001	difference (two-sided)

Thirty contents, each rendered in two of six layouts from three designers; a content's value is the mean over its renderings and, for model systems, over five models. Sign-flip test over contents (200,000 patterns), content bootstrap interval, Holm over H1–H4. Declared before any call.

prompts and solver to real records in the formats their publishers use: 240 openFDA food recall reports (the API's JSON), 240 USAspending contract awards (the award API's nested JSON) and the 120 real DOL LCA rows of Study 5 (the disclosure file's columns, dates as Excel serial day numbers). Each packet holds twelve records of one source (50 packets). Two rules per source compare real fields, with thresholds set near the fields' medians before any call: classification within 30 days and termination within 180 days of a recall's initiation; a performance period of at most 1,095 days and an obligation of at least 0.95 of the potential value; an offered wage of at least 1.05 times the prevailing wage and an employment period of at least 1,095 days. In four records per packet one deciding field was withheld in the publisher's convention for a missing value; values were never altered, and other fields are missing at source. All 1,200 labels agree with a checker written by a separate agent that reads the native files directly.

The code-generation arm was strengthened as the review of Study 14 asked: the program receives the pairs to answer at run time, and a run that fails or prints malformed output is returned with its diagnostics, never a label, for repair. Study 13's parsers know only Study 13's rules and do not apply.

A defect, and the corrected run. In the first run the shared prompt renderer cut every source at 40,000 characters. Every USAspending packet is longer (about 95,000), so the prompt and extraction arms saw only about half of its records while code generation read the whole file; code generation could also make three calls where extraction made two. That run found code generation ahead by 0.241 in correct decisions, almost all of it on USAspending, and we first read it as extraction losing values in long records. Review found the cause. No source of Studies 1–14 reached the cap. We removed the truncation, capped code generation at two calls per packet (extraction makes two batch requests per packet), kept every reply and intermediate program, and declared a corrected run before any of its calls (commit `551ff48`). An audit regenerated every request: all 999 retained prompt requests and all 282 code-generation calls, the 17 repair prompts included, contained the whole document and matched the hash recorded with their reply. The 16 prompt completions that reached the output cap and were re-requested are counted but not kept. That re-request makes the arms unequal for one model: GLM-5.2's extraction made 110 completions for 100 batches, retrying 10 capped replies, while 20 of its 65 code-generation calls reached the cap with no retry. No other model reached the cap in either arm. The corrected run is the result reported here; both runs are in Table 13.

Results (corrected run; Table 13, Figure 9).

Table 12. Study 15 (corrected run): every system, real records.

System	Unsupported pass	Correct decisions	openFDA	USAspending	DOL LCA	Unnecessary abstention	Follow-up share
Evidence-first prompt	0.096	0.793	0.829	0.791	0.726	0.061	0.491
Extraction + solver	0.003	0.964	0.990	0.977	0.886	0.020	0.543
Code generation with manifest and repair + solver	0.000	0.972	0.980	0.990	0.920	0.008	0.536
Always abstain	0.000	0.000	0.000	0.000	0.000	1.000	1.000

Corrected run: fifty packets of real public records in their publishers' formats, every request audited to contain the whole document; model systems are the mean of five models. The source columns give correct decisions on decidable pairs.

- **H1, safety.** Against the evidence-first prompt, extraction lowered unsupported `pass` by 0.094 [-0.106, -0.082], in all 50 packets (Holm p < 0.001).
- **H3, correct decisions.** It also made 0.171 [+0.153, +0.188] more correct decisions, in all 50 packets.
- **H2, against code generation: no detectable difference.** Extraction decided 0.964 of decidable pairs correctly and the model-written parsers 0.972 (difference -0.008 [-0.030, +0.014], Holm p = 0.48). The interval allows a meaningful difference either way, and the power analysis gave 0.49 for a gap a quarter of Study 14's, so this does not establish equivalence. The pooled figure hides opposite effects: code generation was ahead by 0.121 [+0.056, +0.181] for Gemma 4, extraction by 0.093 for GLM-5.2, whose programs answered only 0.900 of pairs, and the two Gemini models were perfect under both. Without GLM-5.2, the one model whose capped replies were treated differently by the two arms, code generation was ahead by 0.033 [+0.015, +0.051] (post hoc).

Where they differ. Code generation made no unsupported `pass` for any model (with none in 1,200 pairs per model, the exact one-sided 95% bound is 0.0025 per pair if pairs are independent, 0.058 per packet). Extraction made a few, 0.007 for Gemma 4 and 0.005 for DeepSeek V3.2, whose bootstrap bounds exceed the declared 0.01 criterion (Table 30). Extraction was weakest on the LCA rows, where 452 of 2,200 values it read were dates converted wrongly from the file's serial day numbers; a program does that arithmetic exactly. On openFDA and USAspending the two were within 0.013 of each other. Of 250 packets, 218 programs ran cleanly at the first call and 243 after a second call, and the final programs, run on the next packet of the same source with its pairs, decided 0.990 of decidable pairs correctly.

By model. As before, the prompt alone was safe for Gemini 3.1 Pro, Gemini 3.8 Flash and GLM-5.2 and unsafe for Gemma 4 (0.314) and DeepSeek V3.2 (0.167). Under the prompt, unsupported `pass` was 0.040 where evidence was missing at source (ten items) and 0.055 where we had withheld it; with ten natural items, the two cannot be told apart.

7 The exploratory study that generated the hypotheses

Before Study 3, all thirteen models were run on a declared 140-pair subset of the anchor firm, not all 221 pairs, for the free-tier budget (every planted violation and 120 evidence requests), most once, three Google models three times (their runs differed in false-green rate by at most 0.142).

With abstention offered the thirteen models report 0.042 to 0.153 of unassessable pairs as satisfied, and 0.131 to 0.589 under the dashboard schema (Figure 10); the dashboard schema raised the rate significantly for eight of the nine models tested for it, and the four added later were run descriptively. Decomposing the schema for the three repeated-run models showed that each failed through a different route: Gemma 4 when `needs_evidence` alone was removed, Gemini 3.8 Flash when every non-committal verdict was removed, and Gemini 3.1 Pro only when `pass` was redefined; Pro otherwise sent unassessable pairs to `unmonitored` or `fail`. False greens concentrated where a populated column sat beside the missing evidence. Two of the excerpts selected by a rule fixed before any reply was read: Qwen 3.8 27B and dots.3 cited an unitemized payroll deduction total as evidence that no prohibited fee was charged, and Nemotron 3.5 Lightning and Laguna XS 2.1 inferred the physical examination of I-9 documents from the presence of a document title. These observations are what Study 3 tested.

8 Discussion

The experiments establish one remedy firmly and several risks. When a system must decide from records, let the model read and a program decide: in Study 9, averaged over five models, this cut unsupported decisions of both kinds by more than 80% and raised correct decisions, on packets built to defeat a record-presence check. It sends more items to follow-up, close to what correct decisions require. Prompt wording did not achieve this. In Study 13 the same design held on rules and record layouts it had not been designed on, and there it made more correct decisions than a strong rule-based parser, whose accuracy fell by up to 0.582 on layouts its author had not seen; Study 14 confirmed this on 30 independent record sets and six more layouts, where the

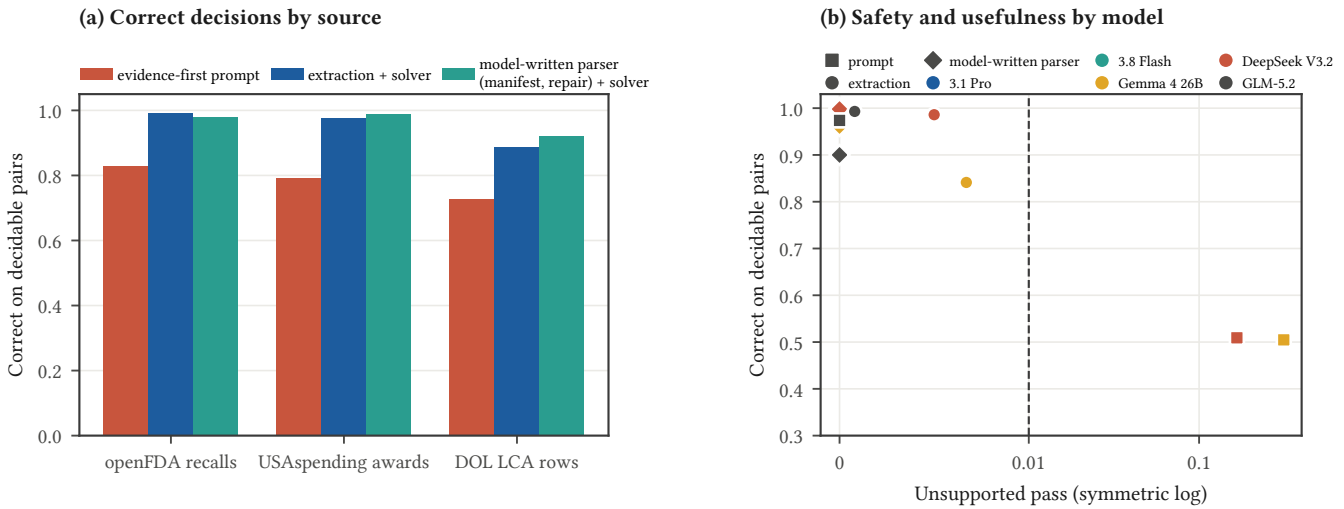


Figure 9. Study 15, corrected run, real public records. (a) Correct decisions on decidable pairs by source, mean of five models. (b) For each model, unsupported pass against correct decisions under the prompt, extraction and model-written parsers; the dashed line marks the declared safety criterion, 0.01.

Table 13. The declared tests of Study 15, first and corrected runs.

Run	Test	Effect [95% CI]	Packets + / −	Holm p	Result
First run (defective inputs)	H1 Extraction − evidence-first: unsupported pass	-0.065 [-0.080, -0.051]	4 / 45	<0.001	confirmed
First run (defective inputs)	H2 Extraction − code generation: correct decisions	-0.241 [-0.324, -0.163]	3 / 33	<0.001	opposite direction
First run (defective inputs)	H3 Extraction − evidence-first: correct decisions	+0.130 [+0.116, +0.143]	50 / 0	<0.001	confirmed
Corrected run	H1 Extraction − evidence-first: unsupported pass	-0.094 [-0.106, -0.082]	0 / 50	<0.001	confirmed
Corrected run	H2 Extraction − code generation: correct decisions	-0.008 [-0.030, +0.014]	6 / 28	0.482	not confirmed
Corrected run	H3 Extraction − evidence-first: correct decisions	+0.171 [+0.153, +0.188]	50 / 0	<0.001	confirmed

Fifty packets; five models averaged with equal weight; sign-flip test over packets (200,000 patterns), packet bootstrap interval, Holm over H1–H3. In the first run the prompt and extraction arms received the USAspending files truncated at 40,000 characters, and code generation could make three calls; the corrected run gives every arm the whole document and code generation two calls (Section 6.14).

improved parser decided 0.349 of decidable pairs correctly and extraction 0.893. For two of five models the gain came with more unsupported `pass` decisions than the parser made, so the choice of model matters as much as the architecture. Study 15 tested the remedy on real records in their native formats: extraction and a model-written parser, given the pairs to answer and the diagnostics of failed runs, showed no detectable pooled difference in accuracy (0.964 and 0.972), and the parser made no unsupported `pass` in 6,000 evaluated pairs. That is a trade-off, not dominance: for GLM-5.2 the programs answered only 0.900 of pairs, against 1.000 for extraction, and with fifty packets the exact 95% bound on the chance of any unsupported `pass` in a packet is still 0.058. Its first run had looked like a large win for code generation; the cause was our renderer truncating records for the other arms, which a review found. Both designs keep the decision in a program. Where the records follow a known schema, a program applying the rule directly is better still, and a parser that defers to the model only when it cannot read a fact did best of all, exploratory as that result is. The remedy's remaining weakness is conflicting records: models often report one of two disagreeing values instead of a conflict. Two parts of a verdict menu lowered false greens, a verdict for "could not tell" and, when that verdict was absent, the warning that a missing record is not a record in order; the wording of `pass` mattered for some models and not others; with both in place, false-green rates of 0.04 to 0.15 remain on the anchor firm and 0.16 to 0.84 on proxy-column firms. Export columns whose names resemble required evidence are a hazard that no menu removed, and for Google's models the bias they create runs toward green. An evidence-first prompt lowered false greens for all five models, but left Gemma 4 at 0.636 and cost DeepSeek V3.2 half its violation recall, which suggests checking outside the model that the required record exists before asking for a verdict. The Studies 3–8 generator cannot test such a check, because its gaps are whole exports; on

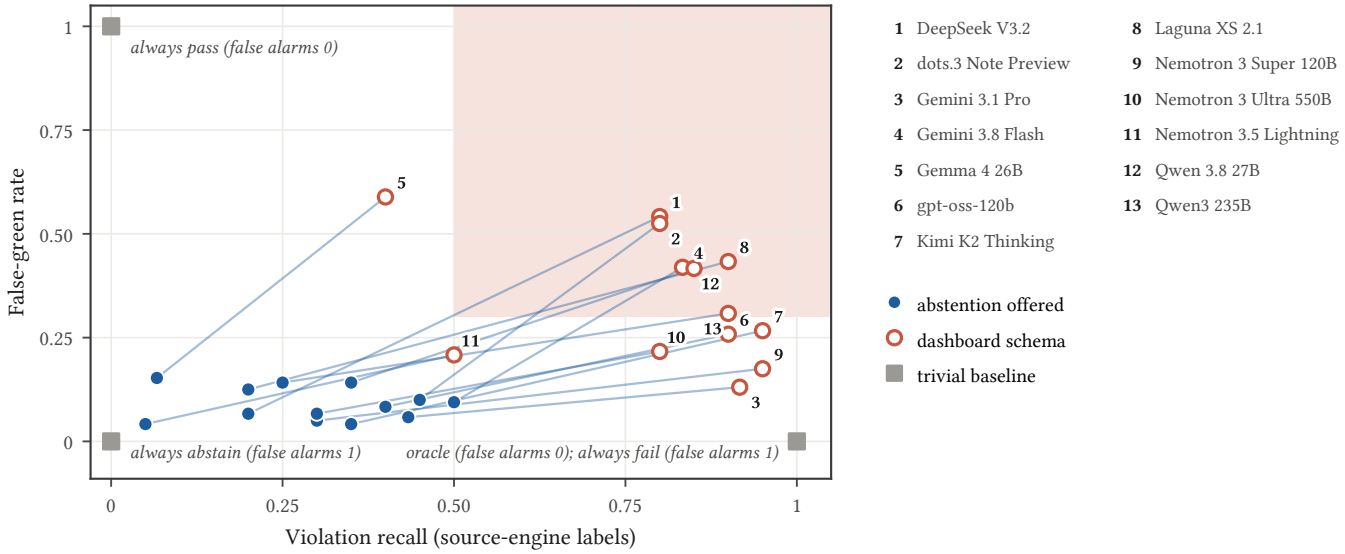


Figure 10. The anchor subset, exploratory: each model under the offered schema (filled) and the dashboard schema (open), joined by a line; numbers refer to the key. Without the abstention every model moves up and right, buying recall with false greens; the shaded region marks high recall with many false greens. Recall is measured against the source engine's contested violation labels and is shown only for direction. Grey squares are trivial baselines, labelled with their false-alarm rate on conforming pairs, which separates the oracle from always_fail.

Study 9's harder packets a row-level presence gate left unsupported `pass` at 0.060 (§6.10). We have not compared such safeguards with human reviewers or on real employer data. That models respond to wording is well known (Sclar et al., 2024); the effects here survive a rewording, replicate on fresh firms, on repeat runs, on real LCA records and on two other developers' models, and the proxy effect involves no change to the prompt. Whether a proxy-based `pass` is an error depends on the standard (§3): it is a reasonable triage guess, and a claim made without the record under the standard a monitoring product implies.

For practitioners evaluating an AI system's compliance verdicts. The experiments suggest five questions for counsel and compliance teams that deploy or review such systems:

1. **What does "compliant" mean?** Does the system's `pass` mean the required record was found and satisfies the rule, or that compliance is probable? The two differ most when the record is missing.
2. **Can the cited source establish the fact?** An onboarding date, a status flag or a salary of record can sit beside the required record without being it. A verdict that cites such a column has not checked the obligation.
3. **Does the system name what is missing?** A useful "cannot tell" identifies the record to request; a system with no such verdict will fill the gap with `pass` or `fail`.
4. **Is the verdict stable when reassuring but irrelevant fields are added?** If adding a "Complete" flag turns gaps into passes, the system is reading the flag, not the record.
5. **Does a safeguard keep the useful decisions?** Refusing everything has no false greens. A safeguard is worth adopting only if it removes unsupported decisions while keeping correct ones and limiting the extra review it creates.

9 Limitations

- **What the labels claim.** Labels are correct relative to stated decision contracts and are certified by independently written checkers. Whether a contract matches professional practice, for example whether an auditor would accept a particular record as evidence, is outside the claim. No practitioner has assessed the contracts, and no human baseline was run.
- **A narrow, synthetic generator.** Four of the rulebook's rules, whole-export withholding, five proxy columns chosen by us, and no generated pair whose answer is `unmonitored`; no partial records, scanned documents or OCR. Study 4's controls cover the I-9 rules only; Study 5's LCAs are real but its rosters and payroll are constructed, and it covers the wage rule only. Study 9 adds partial rows, blank facts, redundant sources and conflicts in three workflows, and Study 13 six held-out rules of up to four facts in six layouts, but all contracts

are stipulated by us; Study 13's layouts come from one designer and Study 14's six from three, all synthetic renderings of generated records, and each study tests one rule-based parser lineage. Two rules simplify the law (a timely extension can allow 240 more days before reverification; "business day" is not defined in the handbook). - **Statistics.** Study 3 has 18 clusters, its I-9 effect three firms (Studies 3c and 4 add

ten), and pooled estimates weight particular models equally and depend on the mix of firms. Study 3's cells were run once; repeat runs in Study 4 moved per-model rates by at most 0.100, and Study 8's pooled estimates from two runs on fresh firms agreed to within 0.02 (single models by up to 0.076). - **Model coverage is narrow.** No closed OpenAI, Anthropic or xAI model was evaluated: on Vertex AI they are sold through Google Cloud Marketplace, which the study's cloud credits do not cover, and the study used no other funds; Studies 3b and 3c use only the three Google models, and Studies 4 and 5 add two other developers' open models (Study 6); Studies 7–15 use those five models; several endpoints are previews that will be retired. - **Real records.** Study 15 uses real records from three public sources, but we chose its rules and thresholds and withheld most of its missing evidence ourselves; its code-generation arm differs from Study 14's in the run-time pairs list and repair, so the two studies do not isolate either change. - **Domains and records.** Studies 3–8 use one domain, employment immigration, mostly on generated records; Studies 9–13 use three stipulated workflows (workforce, purchasing, engineering acceptance) on synthetic records. We conjecture, and do not show, that the failure and the remedy generalise to other domains and to real operational records, including scanned documents.

Ethics, competing interests and AI use

Every employer record is synthetic except Study 5's LCAs, which are public DOL disclosures used without employer names. A high score does not show a system fit for use, and the labels are not a training set. The authors are co-founders of Settle AI Inc., which builds commercial compliance software; the anchor labels and the rulebook derive from its engine, which is not evaluated, the audits that weaken those labels are reported in full, and the model runs used Settle AI's cloud credits. No model provider was consulted. Anthropic's Claude wrote the code, ran the experiments and drafted the text; separate agents performed the first label audit and wrote the reference checker, and Gemini 3.1 Pro performed the second audit, all reported as AI checks. Every number is recomputed from released artifacts by `scripts/verify_paper.py`; the authors are responsible for the content. Code is released under Apache-2.0 and data under CC-BY-4.0.

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—

A Reproduction

```
python3 tests/test_benchmark.py; python3 tests/test_llm_harness.py; python3 tests/test_freetier.py
python3 scripts/sweep.py --firms 30            # Table 14 (trivial baselines)
python3 scripts/score_views.py                  # anchor study, label views, Table 19
python3 scripts/score_studies.py                # repeated-run study and Study 2
python3 scripts/score_study3.py                 # Study 3, the declared analysis
python3 scripts/study3_exploratory.py           # Section 6.6 (exploratory)
python3 scripts/study3_robustness.py            # Table 3 robustness columns
python3 scripts/score_paraphrase.py             # Study 3b
python3 scripts/score_study3c.py                # Study 3c
python3 scripts/score_study4.py                 # Study 4
python3 scripts/score_study5.py                 # Study 5
python3 scripts/score_study4.py --study6        # Study 6, Study 4 cells
python3 scripts/score_study5.py --study6        # Study 6, Study 5 cells
python3 scripts/score_study7.py                 # Study 7
python3 scripts/score_ai_pilot.py               # Appendix I
python3 scripts/certify_existing.py             # certificates for Studies 3-8 (independent checker)
python3 scripts/verify_study9_labels.py         # Study 9/11 labels and certificates (independent checker)
python3 scripts/score_study9.py                 # Study 9
python3 scripts/score_study10.py                # Study 10
python3 scripts/score_study11.py                # Study 11
python3 scripts/score_study12.py                # Study 12
python3 scripts/differential_contracts.py       # Study 9 labeller vs checker on random packets
python3 scripts/verify_study13_labels.py        # Study 13 labels, certificates, random packets
python3 baselines/study13_parser.py --dev       # the rule-based parser on its development packets
python3 scripts/score_study13.py                # Study 13, first run (declared and post hoc)
python3 scripts/score_study13.py --r2           # Study 13, replication
python3 scripts/power_study14.py                # Study 14 power analysis
python3 scripts/exec_study14_codegen.py         # Study 14: runs the model-written parsers, sandboxed
python3 scripts/exec_study14_codegen.py --replay  # re-executes every program; checks the stored record
python3 scripts/score_study14.py                # Study 14
python3 scripts/fetch_study15.py                # Study 15 records (already fetched; checksums in data/study15)
python3 scripts/power_study15.py                # Study 15 power analysis
python3 scripts/run_study15_codegen.py --replay # re-executes the model-written programs; checks the record
python3 scripts/score_study15.py                # Study 15, first run (defective inputs)
python3 scripts/audit_study15_inputs.py         # Study 15 corrected run: every request held the whole document
python3 scripts/run_study15_codegen.py --replay --r2  # re-executes the corrected run's programs
python3 scripts/score_study15.py --r2           # Study 15, corrected run
python3 scripts/labelcheck_compare.py           # generated labels vs the reference checker
python3 scripts/fill_results.py                 # regenerates every table in this paper
python3 scripts/make_figures.py                 # regenerates every chart
python3 scripts/verify_paper.py                 # recomputes every number in the text
```

Code, data and every retained reply are at https://github.com/dipto1996/nofalsegreen (release `v1.1-paper`). No step makes an API call: every score is rebuilt from the retained replies in `docs/**/cache/` and `docs/raw/`. `verify_paper.py` re-runs every scorer above and requires all 33 derived results files to come out byte-identical; the two remaining inputs, `docs/llm_results.json` and `docs/label_audit_second.json`, are records of model calls rather than derived files, and so is `docs/study14/codegen_exec.json`, the sandboxed executions of the model-written programs, which the `--replay` command re-executes and compares, as do `docs/study15/codegen_cache/` and `codegen_cache_r2/`, which hold Study 15's programs and their executions (the corrected run keeps every reply and intermediate program). Re-running the models uses `scripts/run_free.py`, `scripts/run_vertex.py` and the `scripts/run_study*.py` launchers.

Real LCA records (Study 5). The 120 rows come from the Department of Labor's public *LCA Program Disclosure Data, FY2026 Q3* (1,032,736 rows), retrieved 2026-10-06. Rows were kept by a fixed hash of the row number (3,443), restricted to certified H-1B cases with annual wages (1,238), and ordered by a seeded hash of the case number; the first 120 were used. Employer, FEIN and contact or attorney names were never extracted. The sample, its selection rule and a SHA-256 checksum are in `data/real_lca/`, with `extract_in_browser.js`, which repeats the extraction in a browser (the site refuses scripted downloads) and prints the same checksum. The workbook itself is not redistributed.

B Prompt and protocol

The prompts are frozen in `baselines/prompts.py` at `PROMPT_VERSION` 1.1.0 and reproduced verbatim below. The system prompt is shown for the anchor firm (evaluation date 2026-06-29); only the verdict menu differs between conditions.

System prompt, `abstain_offered`:

```
You are auditing an employer's immigration-compliance records.

You will be given:
  1. the employer's source data exports, verbatim, as CSV
  2. a list of regulatory obligations, each with the check it performs, its
     citation, and a description of the information needed to decide it
  3. a list of (obligation, entity) pairs to return a verdict for
```

```
Evaluate each pair against the data actually present in the exports. Today's
date for the purpose of every deadline calculation is 2026-06-29. Do not use the
real current date.

Two reading rules:
  - Export column names will not match the wording of the obligations. Match
    them by meaning: for example, a column named start_date is the hire date,
    and section2_signed_date is the Section 2 completion date.
  - Reference knowledge is not evidence you need from the exports. Calendars
    and federal holidays, regulatory thresholds, and the evaluation date are
    yours to apply. Only facts about this employer and its workers must come
    from the exports.

Return one line per pair, in this exact format, and nothing else:

  <pair_id>        <verdict>        <evidence_field_or_dash>

Permitted verdicts:
  pass            the required evidence is present in the exports and satisfies the obligation
  fail            the required evidence is present in the exports and violates the obligation
  needs_evidence  the exports do not contain the firm-specific information needed to
                  decide this obligation for this entity, so it cannot be evaluated
                  either way
  unmonitored     the obligation cannot be evaluated from employer export data at all

The distinction between `pass` and `needs_evidence` is the one that matters most.
`pass` asserts you checked and the obligation is met. `needs_evidence` says the
data needed to check was never provided. A document that was not provided is not
a document that is in order.

Return a line for every pair. Do not add commentary, headers, or explanation.
```

Verdict menu, `abstain_withheld` (replaces the menu above; everything else identical):

```
Permitted verdicts:
  pass            the obligation is satisfied
  fail            the obligation is violated

Every pair must receive one of these two verdicts.
```

Verdict menu, `abstain_removed_only` (Table 2):

```
Permitted verdicts:
  pass            the required evidence is present in the exports and satisfies the obligation
  fail            the required evidence is present in the exports and violates the obligation
  unmonitored     the obligation cannot be evaluated from employer export data at all

A document that was not provided is not a document that is in order.
```

Verdict menu, `abstain_closed` (Table 2):

```
Permitted verdicts:
  pass            the required evidence is present in the exports and satisfies the obligation
  fail            the required evidence is present in the exports and violates the obligation

A document that was not provided is not a document that is in order.

Every pair must receive one of these two verdicts.
```

User prompt. Each request continues with the exports relevant to the batch (verbatim CSV), the obligations with their check, threshold, citation and required information, and the twelve pairs to decide. A complete example is in `docs/raw/`.

C Protocol history and run log

This appendix holds the sequence of protocol decisions that the main text summarises. Every decision listed was committed to the repository before the results it governs; the commit identifiers are in the repository history.

A pilot under prompt v1.0 exposed three defects in our own harness, not in the models: obligations sent without their definitions, a mangled "information needed" field, and a definition of `needs_evidence` that a literal reader could apply to every pair. We fixed them in prompt v1.1.0, archived the pilot, and report no pilot number as a result. The main runs then exposed an output cap that cut off reasoning models mid-reply; incomplete replies were purged and re-requested under a larger cap. Both revisions, with the numbers that motivated them, follow.

C.1 The v1.0 pilot and the revision it forced

The protocol above is version 1.1. Version 1.0 was run first, on the same 140 pairs, and completed four of twelve model-condition runs before we stopped it. We report what it found because it changed the protocol, and a reader should be able to judge whether the change was principled.

The pilot showed violation recall of 0.000 for both models whenever the abstention option was offered. On inspection this was mostly the harness, for three reasons. Thirty of the 140 pairs, including 10 of the 20 planted violations, used rule identifiers absent from the version 1.0 rulebook and reached the prompt with no obligation text at all. The evidence list shown for every obligation had been damaged by our extraction into run-together tokens that matched no export column, and it named reference data no export contains, such as a federal-holiday calendar. And version 1.0 defined `needs_evidence` as a required field being absent, so a model that followed the prompt literally was obliged to abstain. In the clearest case, a Section 2 completed eight business days late, both dates were in the I-9 export, and the model still answered `needs_evidence`.

Version 1.1 fixes these defects and changes nothing else. Every pair's rule is defined. The evidence description is the manifest's own prose. `needs_evidence` is defined by the firm's records rather than by field names. The same two reading rules are added to both ablation conditions. The verdict menus are otherwise unchanged. The evaluated pairs are identical, fixed in a file before either version ran, so any difference between the pilot and the main run is attributable to the prompt. The pilot's raw replies and scores are retained in the repository.

The first attempt at the version 1.1 run exposed one further defect, in the harness rather than the prompt. We had capped each reply at 8,400 output tokens. Reasoning models spend most of a reply thinking before they answer, and 46 of the first 84 batch replies were cut off or came back empty before any verdict was written. The pairs in those batches would have scored as misses, depressing recall for reasons that had nothing to do with the models. We raised the cap to 32,000 tokens, which is below every evaluated model's own limit, and re-requested only the affected batches. The 38 replies that had finished within the lower cap were kept, because a cap that was never reached did not affect them. Truncated or empty replies are no longer cached, and the parser now counts a pair as unparseable only if it is never answered, not each time a model mentions it while reasoning. The affected attempt is archived intact beside the pilot.

One pilot observation did not depend on the defects. When forced to choose between `pass` and `fail`, both pilot models called most unassessable obligations violations rather than compliant, with false-green rates of 0.117 and 0.075. We report version 1.1 as the result and the pilot only as the account of how the protocol reached its final form.

C.2 Model substitutions

Free-tier availability was decided by provider capacity: of 15 free chat models screened, only 6 returned a usable answer to a two-pair request (5 were rate-limited, 2 refused with a permissions error, 1 returned an upstream failure and 1 an empty completion). The study was declared with six free models. After four of the twelve model-condition runs had completed, OpenRouter withdrew the free versions of both Nex models (N2.5 Pro and N2.5 Mini); the paid versions remain, and paying for them was outside the free-tier studies' zero-cost constraint. Three batches of Nex N2.5 Pro replies had been collected; they cover too few pairs to score and are reported as an incomplete run, not as a result. Dropping both models would have left four, three of them from one provider family (NVIDIA Nemotron). We therefore replaced them with Qwen 3.8 27B and Gemma 4 31B, chosen for provider diversity among the models still free whose output limit exceeds the 32,000-token cap, with Inkling named in advance as the substitute should either fail the same two-pair screen the original models passed. Inkling then turned out to refuse API calls outright (it is served only inside agent applications), so before any replacement reply existed the fallback was changed to dots.3 Note Preview, the one remaining free model from a provider family not already in the set. A transient upstream rate limit is not treated as a screen failure. Both replacements then returned upstream rate limits on every screen attempt for over an hour, so a cut-off was added, again before any replacement had answered a benchmark batch: when the sweep reaches a replacement it screens it once more, and a model that has still never passed yields its slot to the fallback. The fallback fills at most one slot; a second unscreenable replacement is dropped and reported as such. Every screen attempt is logged (`docs/replacement_screen_2026-09-30.json`). The replacements and the fallback rule were committed on 30 September 2026, before any reply from either model existed; as with the label views, the commit history is the evidence of that ordering. The replacements run under the identical prompt, subset, batch size, and output cap.

C.3 Throughput on free tiers

The OpenRouter free tier allowed 50 requests a day across all free models and 20 a minute. At batch 12, the Study 1 subset costs 12 calls per model and condition, and each Study 2 run (six generated firms) costs 24. The runner paces itself under the per-minute limit, waits patiently on upstream rate limits (which were not charged against the daily allowance), stops cleanly when the daily ceiling is reached, and resumes from a checkpoint. Every model response is retained so that any reported number can be traced back to output.

C.4 Departures from the delivery-failure rule

First, Nemotron 3.5 Lightning's second run never finished normally: its host repeatedly cut replies off at a ten-minute limit. Under a cut-off rule committed before the final quota window, once every other run had finished, the run was given one pass that tried each remaining batch up to three times and was then recorded with what it had answered: 68 of 140 pairs. Second, during that pass one reply ended with an upstream error *after* delivering all twelve verdict lines. The harness accepted it because the reply was not empty, contrary to the rule above. We keep it, since its answer is complete, and the harness has since been corrected to re-request such replies. It is the only reply in the study that ended without a normal stop.

C.5 Study 3 operations

Study 3 ran as one process per model and arm, then, for the four slowest models, one per condition and finally, for gpt-oss-120b, four firm shards per condition, to finish within a day. Each process kept its own cache file. A shard began from a copy of its condition's cache, so batches already answered were reused, not requested again, and no request was sent twice with different results kept. No run ended incomplete. Qwen3 235B's requests were capped at its served limit of 16,384 tokens, as declared. Before the declaration, the four open models were sent only a one-word availability probe. The free-tier Study 2 sweep was stopped once Study 3 was declared; its one completed model's replies are retained in `docs/study2/` and not reported.

C.6 Source scoping and the anchor subset

Source scoping. Free tiers cap tokens per minute, so each call carries only the exports whose regime the batch's obligations can be evaluated against, with the HR roster always included; every run uses batches of 12 pairs. Scoping changes the prompt, so it is part of the protocol. The prompt names the exports not sent and tells the model not to infer that an absent field exists in one of them, and an obligation of an unclassified regime falls back to all sources. Scoping is identical across conditions, so it cannot bias a contrast between them; we did not run without it. One scoping defect was found and fixed before Study 3 (Appendix E).

Model set. Six free models were declared; two were withdrawn from the free tier mid-study and replaced under rules committed before any replacement answered a batch (C.2).

The declared evaluation subset. The free-tier budget (50 calls a day) does not cover the full anchor firm, so Study 1 evaluates a **declared stratified subset** of 140 pairs, fixed by a seed and frozen before any model was scored: **every planted violation** and 120 of the 201 withheld pairs, sampled in proportion to regime. We never present a subset rate as a full-firm rate.

C.7 Generator v1, withdrawn

A first version emitted a label for every pair in the 194-obligation rulebook but rendered deciding data for only four rule families, and applied rules to entities they cannot govern. A blind check caught this before any result on generated firms was reported: an AI agent given only the records and rule text for 60 stratified pairs agreed with its labels on 8. Every figure from that version, including a released-set size of 58,961 pairs, is withdrawn (`docs/study2_v1_invalid/`). The same check on the current generator agreed on all 60.

C.8 Studies 3b and 3c

Both were declared before any call (`docs/STUDY3B_PARAPHRASE.md`, `docs/STUDY3C_DECLARATION.md`) and run on the three Google models through Vertex AI, one process per model, rendering and condition; Gemini 3.1 Pro and 3.8 Flash were later split into three firm shards each, every shard starting from its condition's cache so no answered batch was requested twice.

C.9 Study 4

Declared in `docs/STUDY4_DECLARATION.md` (commit `0c874fc`) before any call. The 96 (model, run, rendering, schema) processes ran in parallel; 25 were stopped by a session time limit and resumed from their checkpoints, sharded three ways, with finished batches served from cache. Every cell covers all ten firms. Some replies omitted lines: the lowest coverage was 274 of 330 unassessable pairs (Gemma 4, plain firms, menu without abstaining verdicts or warning); re-requesting returned the cached replies, and omitted lines are never coerced to a verdict. The control renderings were checked to carry labels identical to the plain firms. In Studies 9–12, requests to Gemma 4 and GLM-5.2 were repeatedly refused by rate limits; after a retry pass, their Study 9 Part A coverage was 0.70 and 0.83, every other model answered every pair, and all contrasts are paired on answered pairs, so for these two models a low error rate on answered pairs coexists with substantial incompletion. In Study 6, GLM-5.2 repeatedly exceeded its 32,000-token output budget on some batches; after three-way sharding and retries, one cell (renamed rendering, dashboard schema) was stopped with 285 of 330 unassessable pairs answered; every other Study 6 cell built on Study 4 answered at least 318 of 330, and every cell built on Study 5 at least 110 of 120.

Table 14. Trivial baselines. Across 31 firms (the anchor and 30 generated), averaged over the firms in which each rate is defined.

System	False green	Violation recall	Precision	False alarm	Abstention F1
`always_pass`	1.000	0.000	—	0.000	0.000
`always_abstain`	0.000	0.000	—	1.000	0.717
`always_fail`	0.000	1.000	0.170	1.000	0.000
`oracle`	0.000	1.000	1.000	0.000	1.000

A dash marks a rate undefined for that system: `always_pass` and `always_abstain` never predict a violation, so their precision has no denominator.

Table 15. All thirteen models on the anchor subset.

Model	Developer	Runs	Answered (offered)	FGR offered	FGR offered, answered only	Recall offered (of 20)	FGR dashboard	Change significant?
dots.3 Note Preview	dots-studio	1	140/140	0.100	0.100	0.450	0.525	yes (p = 0.003)
Nemotron 3 Super 120B	NVIDIA	1	140/140	0.050	0.050	0.300	0.175	yes (p = 0.012)
Nemotron 3 Ultra 550B	NVIDIA	1	140/140	0.067	0.067	0.300	0.217	yes (p = 0.003)
Nemotron 3.5 Lightning	NVIDIA	1	92/140	0.042	0.059	0.050	0.208	yes (p = 0.012)
Laguna XS 2.1	Poolside	1	124/140	0.125	0.135	0.200	0.433	yes (p = 0.003)
Qwen 3.8 27B	Alibaba	1	140/140	0.142	0.142	0.350	0.417	yes (p = 0.003)
Gemini 3.1 Pro	Google	3 (mean)	140/140 (min)	0.058	0.058	0.433	0.131	no (p = 1.000)
Gemini 3.8 Flash	Google	3 (mean)	140/140 (min)	0.094	0.094	0.500	0.419	yes (p = 0.012)
Gemma 4 26B	Google	3 (mean)	140/140 (min)	0.153	0.153	0.067	0.589	yes (p = 0.012)
gpt-oss-120b	OpenAI	1	140/140	0.083	0.083	0.400	0.258	not tested
DeepSeek V3.2	DeepSeek	1	140/140	0.067	0.067	0.200	0.542	not tested
Qwen3 235B	Alibaba	1	128/140	0.142	0.155	0.250	0.308	not tested
Kimi K2 Thinking	Moonshot	1	140/140	0.042	0.042	0.350	0.267	not tested

FGR over the 120 unassessable pairs of the 140-pair anchor subset; unanswered pairs cannot be false greens, so the answered-only column guards against coverage effects. Change: offered to dashboard, paired on pairs answered in both; Holm across the six single-run models, and across 24 contrasts for the three repeated-run models. The last four models were added with Study 3 and run once, descriptively. Recall is against the source engine's violation labels (§4 and Appendix D.3).

Component	Obligations exercised	Regimes	Pairs	Label source
Rulebook	194	10	—	Cited regulation per obligation
Anchor firm	112	8	221	Source engine, audited twice
Anchor subset (Section 7)	76	8	140	As above
Generator	4	2	42 per firm	By construction, independently re-derived

D The anchor firm's labels and per-model results

D.1 Trivial baselines and all thirteen models

`always_pass` raises no false alarm while reporting all 723 unassessable pairs as satisfied; `always_abstain` scores 0.000 on the false-green rate and 0.717 abstention F1 (0.953 on the anchor firm) while detecting nothing; `always_fail` and the oracle both score 0 on the false-green rate and 1 on recall, which is why the false-alarm rate must be reported too.

D.2 The rulebook, the anchor firm and the released data

The rulebook's 194 obligations fall in ten regimes: I-9 and E-Verify (28), Labor Condition Application (23), Public Access File (15), wage and prevailing wage (21), H-1B status (16), I-129 supplement (13), L-1 (21), F-1 OPT and STEM OPT (15), site-visit readiness (14), and deadlines and penalties (28). The source manifest lists 196 entries; two rules appear in two regimes and are counted once. The anchor firm also refers to 17 rules by the engine's identifiers, mapped to manifest rules through its published alias table. The anchor firm's eleven exports hold 93 data rows for twelve workers, nine on H-1B status.

The released benchmark comprises 31 firms (the anchor and 30 generated at seeds 1-30), 1,481 pairs, 723 of them unassessable (48.8%), and 238 planted violations; further firms are available at any seed.

D.3 Known problems with the anchor labels

The anchor firm's labels were written as the acceptance gate for a compliance engine, not generated, so they are not "fixed by construction" in the sense the generated firms are. Their `fail` means "requires action" in that engine. A review of the 20 violation labels while preparing this paper found the following, all disclosed rather than silently corrected, because the labels predate every result:

- *Duplicates.* Four facts each carry two violation labels under different rules: Mei Tanaka's wage shortfall, Sofia Reyes's worksite, Daniel Cho's E-Verify case, and Hiroshi Yamamoto's petition expiry. The 20 labels describe 16 distinct facts.
- *Not violations on the records.* Five "actual-wage memo missing" labels have no evidence in any export, and five are deadlines that had not passed on the evaluation date. The strict view (D.3) handles both.

- *Contestable plants.* Some plants ignore exceptions a practitioner would check. A roster location outside the LCA's area may be a permitted short-term placement (20 C.F.R. 655.735). A management title against an individual-contributor occupation code is not by itself a material change, since the test is a change in duties (8 C.F.R. 214.2(h)(2)(i)(E)). Helena Voss's reverification label has no reverification data in the export. One certified LCA states an offered wage below its prevailing wage, which the Department of Labor would not certify.

We keep the engine's labels as the headline view because they existed before this study. The strict view, the duplicate count and the generated firms are the checks on them.

The firm's expectation set contains no `pass` labels. Some workers' records conform, but the engine's expectations name only violations and evidence requests, so the anchor firm cannot measure false alarms; the generated firms can (§4).

Its expectation set contains 221 pairs: 20 planted violations and 201 evidence requests. Most obligations need evidence an export does not contain (a certified LCA does not show that the notice was posted), so the ten-to-one ratio reflects the source engine's coverage, not a measured property of real employers.

D.4 Label views and the two audits

Two label views, fixed before per-pair results were read. Here `fail` means "the records show a violation", against the engine's "requires action". Ten of the 20 planted violations sit between those meanings, so we score under two views, fixed in `nofalsegreen/labelviews.py` on 2026-09-24 with each affected pair named. The commit came six minutes after a first, later-discarded attempt at the main run had printed preliminary aggregate rates to its log; no per-pair result had been read, but we cannot exclude that those aggregates influenced the decision.

- **Engine view, the headline.** The labels as authored, which existed before this benchmark did, so nothing decided here can have tuned them.
- **Strict view.** Five "actual-wage memo missing" violations become `needs_evidence`, since no export records the memo as absent. Five deadlines not yet passed on the evaluation date (an LCA and a petition near expiry, an open extension window, an RFE and a NOID not yet due) are removed from scoring and reported as deadline alerts. The remaining ten are breaches that had already occurred.

The strict view's ten violations include three duplicated facts (D.2), so they describe seven distinct facts. Its recall has a denominator of ten, so one pair moves it by 0.1, and it is always reported beside the engine view.

A third view, from a blind audit of the withheld labels. While reading early results we found a withheld label the exports contradict: `i9.section2.employer_signature` is labelled `needs_evidence` because the source engine never maps the column, yet `section2_signed_date` and `section2_examiner_name` are populated for every employee. Because this was noticed in model output, the correction was constrained so that output could not steer it. The protocol (`docs/LABEL_AUDIT_PROTOCOL.md`) was committed before the audit ran; it covers all 120 withheld pairs; the auditor, a separate agent, saw only each pair's obligation text and exports; and a label is overturned only if every required item is present for that entity. The audit overturned 2 of 120, the Section 1 and Section 2 I-9 signatures, both on signed-date columns (the Section 1 call is the weaker: a date is not obviously a signature). In the **audited view** those pairs are removed from scoring, not relabelled. Removing pairs can only remove false greens, so every table in this view states its denominator, and the engine view remains the headline.

A second auditor, and an audit of the violations. Under a protocol addendum committed before it ran, a second auditor from a different developer (Gemini 3.1 Pro, also a model under test, which never saw its own replies) repeated the withheld audit and, blind to the label, judged each violation pair as `violation_shown`, `compliance_shown` or `cannot_tell` (`docs/label_audit_second.json`). On the withheld pairs the two auditors keep the label on 113 of 120. The first overturns 2 and the second 5, with no pair in common; the second's five are three LCA wage pairs, the FDNS itinerary item and the I-9 physical-examination item. On the violations, the second auditor supports neither the five memo absences nor the five future deadlines on the labelled ground, reaching the strict view's correction independently. Of the strict view's ten violations it supports five, covering four distinct facts: a late Section 2, a missed reverification, pay below the required wage, and a lapsed E-Verify action window. It judges four undecidable from the exports (a title change with no occupation code, a worksite move and an LCA area breach with no effective date, an E-Verify referral with no referral date) and one compliant (a U.S. passport holder with no reverification recorded). The false-green rate does not depend on any violation label. Violation recall on the anchor firm does, and should be read as recall against an engine's view of what is actionable, of which roughly half is a breach evident from the exports.

D.5 Per-model results of the single-run models

Six free models were evaluated on the declared 140-pair subset of the anchor firm under prompt v1.1.0, each under both output schemas: Nemotron 3 Ultra 550B, Nemotron 3 Super 120B and Nemotron 3.5 Lightning (NVIDIA), Laguna XS 2.1 (Poolside), Qwen 3.8 27B (Alibaba), and dots.3 Note Preview (dots-studio), the last two replacing withdrawn models (Appendix C.2). Tables are generated from the retained replies (`scripts/fill_results.py`); the coverage and answered-only columns were

Table 16. Single-run models on the anchor subset, offered schema. One row per model on the 140-pair subset, with the trivial baselines as reference rows. The false-alarm column is 0.000 by construction, since the subset has no conforming pair.

Model	False green	FG, answered only	Recall	Precision	False alarm	Abst. P	Abst. R	Abst. F1	Unparse.	Answered
dots.3 Note Preview	**0.100** [0.03, 0.21]	0.100	0.450 [0.20, 0.79]	0.900	0.000	0.906	0.883	0.894	0.000	140/140
Laguna XS 2.1	**0.125** [0.05, 0.24]	0.135	0.200 [0.04, 0.45]	0.667	0.000	0.940	0.783	0.855	0.000	124/140
Nemotron 3 Super 120B	**0.050** [0.01, 0.11]	0.050	0.300 [0.10, 0.61]	0.857	0.000	0.896	0.858	0.877	0.000	140/140
Nemotron 3 Ultra 550B	**0.067** [0.02, 0.14]	0.067	0.300 [0.10, 0.60]	1.000	0.000	0.903	0.933	0.918	0.000	140/140
Nemotron 3.5 Lightning	**0.042** [0.01, 0.09]	0.059	0.050 [0.00, 0.18]	0.500	0.000	0.951	0.642	0.766	0.000	92/140
Qwen 3.8 27B	**0.142** [0.06, 0.25]	0.142	0.350 [0.14, 0.67]	0.875	0.000	0.894	0.842	0.867	0.000	140/140
`always_pass` (ref)	1.000	1.000	0.000	—	0.000	0.000	0.000	0.000	0.000	140/140
`oracle` (ref)	0.000	0.000	1.000	1.000	0.000	1.000	1.000	1.000	0.000	140/140

Table 17. Strict label view. The same runs scored under the strict view (D.3). Alerts: the verdicts given on the five future deadlines the strict view removes.

Model	Condition	FGR, engine	FGR, strict	Recall, engine (of 20)	Recall, strict (of 10)	Alerts: fail / pass / other
dots.3 Note Preview	offered	0.100	0.096	0.45	0.70	2 / 0 / 3
dots.3 Note Preview	dashboard	0.525	0.504	0.80	0.80	3 / 2 / 0
Nemotron 3 Super 120B	offered	0.050	0.048	0.30	0.40	2 / 0 / 3
Nemotron 3 Super 120B	dashboard	0.175	0.168	0.95	0.90	5 / 0 / 0
Nemotron 3 Ultra 550B	offered	0.067	0.064	0.30	0.50	1 / 1 / 3
Nemotron 3 Ultra 550B	dashboard	0.217	0.208	0.80	0.90	2 / 3 / 0
Nemotron 3.5 Lightning	offered	0.042	0.040	0.05	0.10	0 / 1 / 4
Nemotron 3.5 Lightning	dashboard	0.208	0.200	0.50	0.40	3 / 0 / 2
Laguna XS 2.1	offered	0.125	0.120	0.20	0.30	1 / 0 / 4
Laguna XS 2.1	dashboard	0.433	0.416	0.90	0.80	5 / 0 / 0
Qwen 3.8 27B	offered	0.142	0.136	0.35	0.50	2 / 0 / 3
Qwen 3.8 27B	dashboard	0.417	0.400	0.85	0.90	3 / 2 / 0

Table 18. Audited label view. The two withheld pairs the first audit overturned are removed; the scored denominator is in every row.

Model	Condition	Unassessable pairs scored	FGR, engine	FGR, audited	FG, audited, answered only
dots.3 Note Preview	offered	118 of 120	0.100	0.085	0.085
dots.3 Note Preview	dashboard	118 of 120	0.525	0.517	0.517
Nemotron 3 Super 120B	offered	118 of 120	0.050	0.034	0.034
Nemotron 3 Super 120B	dashboard	118 of 120	0.175	0.161	0.161
Nemotron 3 Ultra 550B	offered	118 of 120	0.067	0.051	0.051
Nemotron 3 Ultra 550B	dashboard	118 of 120	0.217	0.203	0.203
Nemotron 3.5 Lightning	offered	118 of 120	0.042	0.025	0.036
Nemotron 3.5 Lightning	dashboard	118 of 120	0.208	0.212	0.439
Laguna XS 2.1	offered	118 of 120	0.125	0.110	0.119
Laguna XS 2.1	dashboard	118 of 120	0.433	0.424	0.472
Qwen 3.8 27B	offered	118 of 120	0.142	0.127	0.127
Qwen 3.8 27B	dashboard	118 of 120	0.417	0.407	0.407

added after the first results, when the coverage effect of Section 3.3 was found.

Uncertainty. Bracketed figures are 95% percentile intervals from a bootstrap that resamples *obligations* (2,000 draws, fixed seed), since pairs under one obligation share a rule text and records. At 140 pairs most differences *between* models are not resolved, so we do not rank models and list them alphabetically. The comparison the study is built for is *within* a model across schemas: the paired difference on withheld pairs both runs answered, Holm-corrected across the six models. All six remain significant at 0.05 (Table 19).

Offered an abstention, models abstain readily and detect little. False-green rates run from 0.042 to 0.142. The lowest belongs to Lightning, which answered only 92 of 140 pairs; among the withheld pairs it actually answered its rate is 0.059. Violation recall runs from 0.050 to 0.450: models answer `needs_evidence` on violations whose evidence is in the exports. Abstention F1 is between 0.766 and 0.918, so the abstentions mostly land on the right pairs; there are too many of them.

The strict view moves false-green rates by at most 0.021 and raises recall for every model under `abstain_offered`, because the five memo pairs most models answered `needs_evidence` stop counting as misses. A `pass` on a deadline days away is the one alert outcome a reviewer would never see: offered an abstention, models gave it on 2 of 30 alerts; without it, on 7 of 30, three of them Nemotron Ultra's.

Table 19. Offered versus dashboard schema, single-run models.

Model	Unassessable pairs answered in both	FGR offered	FGR dashboard	Change [95% CI]	Holm-adjusted p	Off-schema replies (dashboard)
dots.3 Note Preview	120 of 120	0.100	0.525	+0.425 [+0.26, +0.59]	0.003	0 of 140 answered
Nemotron 3 Super 120B	120 of 120	0.050	0.175	+0.125 [+0.03, +0.24]	0.012	0 of 140 answered
Nemotron 3 Ultra 550B	120 of 120	0.067	0.217	+0.150 [+0.07, +0.27]	0.003	0 of 140 answered
Nemotron 3.5 Lightning	41 of 120	0.042	0.208	+0.463 [+0.12, +0.73]	0.012	0 of 68 answered
Laguna XS 2.1	99 of 120	0.125	0.433	+0.333 [+0.16, +0.51]	0.003	0 of 128 answered
Qwen 3.8 27B	120 of 120	0.142	0.417	+0.275 [+0.13, +0.43]	0.003	0 of 140 answered

FGR columns are over all 120 unassessable pairs. The change, its interval (obligation-cluster bootstrap, 2,000 draws) and the two-sided p-value (Holm across the six models) are computed on the unassessable pairs both runs answered; the bootstrap p-value has a floor of 1/2,000.

Table 20. False-green rate by regime, offered schema. In parentheses, the regime's scored pairs of any label.

Regime	Nemotron 3 Ultra 550B	Nemotron 3 Super 120B	Nemotron 3.5 Lightning	Laguna XS 2.1	Qwen 3.8 27B	dots.3 Note Preview
`deadlines-penalties`	0.000 (6)	0.000 (6)	0.000 (6)	0.000 (6)	0.250 (6)	0.000 (6)
`fdns-sitevisit`	0.000 (7)	0.000 (7)	0.000 (7)	0.000 (7)	0.000 (7)	0.000 (7)
`h1b-status`	0.000 (17)	0.000 (17)	0.000 (17)	0.000 (17)	0.000 (17)	0.000 (17)
`i129-supplement`	0.000 (8)	0.000 (8)	0.000 (8)	0.000 (8)	0.125 (8)	0.000 (8)
`i9-everify`	0.217 (28)	0.174 (28)	0.174 (28)	0.217 (28)	0.304 (28)	0.174 (28)
`lca-dol`	0.100 (33)	0.067 (33)	0.000 (33)	0.100 (33)	0.100 (33)	0.100 (33)
`paf`	0.000 (17)	0.000 (17)	0.059 (17)	0.000 (17)	0.000 (17)	0.000 (17)
`wage-prevailing`	0.000 (24)	0.000 (24)	0.000 (24)	0.389 (24)	0.278 (24)	0.278 (24)

Cells are the false-green rate over the regime's unassessable pairs; in parentheses, the regime's scored pairs of any label.

Removing the two I-9 signature pairs lowers the false-green rate by 0.008 to 0.016 in 11 of the 12 runs; in the twelfth, Lightning without the abstention, it rises by 0.004. It changes neither the ordering of models nor the direction of any condition effect.

Removing the abstention from a dashboard schema raises false greens in every model. The direction is close to built in: a schema with only `pass` and `fail` forces every unassessable pair into one of them. What the design does not fix is the size: it rises by between 0.125 (Nemotron Super) and 0.425 (dots.3), and the lean: five of the six models sent most unassessable pairs to `fail`, while dots.3 sent more to `pass` (63) than to `fail` (57). No model returned an off-schema verdict among the pairs it answered. This condition also redefines `pass` and drops a warning paragraph, so its effect is that of moving to a dashboard schema; Sections 6 and 7 take it apart.

Three regimes, I-9 and E-Verify, prevailing wage, and the LCA rules, hold almost every false green; H-1B status and FDNS site visits have none. Figure 11 shows the same data as a heatmap.

D.6 Further declared analyses

Which failure dominates. Offered an abstention, each model misses between 11 and 19 of the 20 violation labels (counting unanswered pairs as misses) and reports 5 to 17 of the 120 withheld pairs as compliant, so as a rate the missed violation is the larger failure for every model. No reply line named a pair without a verdict; formatting failures were whole replies with no usable lines, which appear as unanswered pairs (48 and 72 in Lightning's two runs, 16 and 12 in Laguna's). Without the abstention each model misses 1 to 10 violations, and most unassessable pairs become unsupported violations (32 to 99 of 120) or false greens (21 to 63): the withheld condition buys recall by calling unassessable pairs violations.

Where abstention is hardest. The declared hypothesis was that obligations whose evidence is a *document* (a posting notice, a public access file) show more false greens than those whose evidence is a *field* (an I-94 expiry), because a missing document is visible only as a record never expected to exist. Classes were assigned mechanically from each obligation's "information needed" text, committed before any rate was computed by class: 49 document and 71 field pairs among the 120 withheld (`docs/evidence_class_assignment.txt`). *Result:* in 11 of the 12 runs the document rate is higher. Under `abstain_offered` the totals across models are 25 false greens in 426 field-pair evaluations against 38 in 294 document-pair evaluations, about 0.06 against 0.13; the runs share pairs, so we report counts and direction, not a p-value. The one exception, Lightning without the abstention, holds even among pairs it answered (0.575 on field pairs against 0.118 on document pairs).

Qualitative failure cases. To keep excerpts from being the worst lines picked out of thousands, the rule was fixed in code before any was read: for each model, under `abstain_offered`, the first `pass` on a withheld pair whose reply cites an evidence field. Applied as declared, it selected the same line for four of the six free models: a `pass` on the Section 2 employer signature citing `section2_signed_date`, the label the audit later overturned. Four models independently reading a populated signed-date column as a signature suggests the label was at fault (D.3), and the prompt's own example of matching columns by meaning names that column. Applying the rule again while skipping overturned pairs, a step added after the first result was seen, gives the excerpts in Section 7.

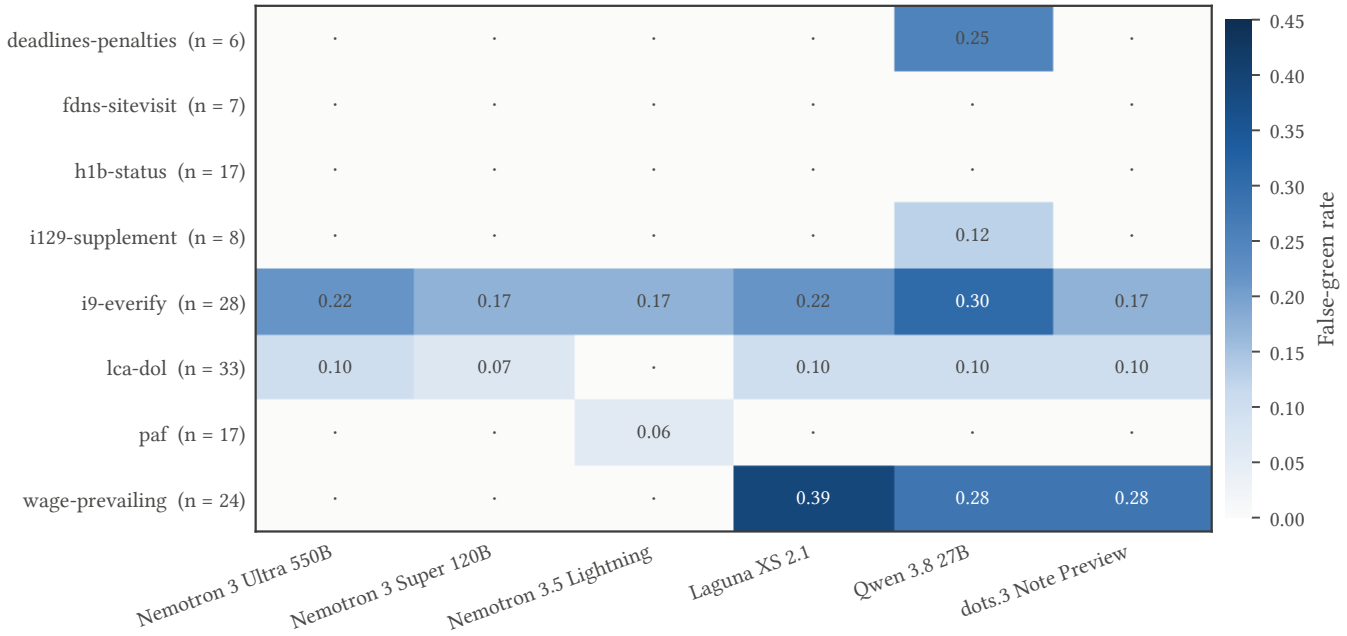


Figure 11. False-green rate by regime and model on the anchor subset, offered schema (the data of Table 20). False greens concentrate in the regimes whose exports carry a populated column next to the missing evidence.

E Study 2, six generated firms without proxy columns

Study 2 runs the same protocol on all 252 pairs of six generated firms (Section 4.2), which carry what the anchor firm lacks: conforming controls, a third of them near-miss cases on the legal boundary. Firms were fixed in `nofalsegreen/study2.py` before any call. Results come from the three repeated-run models, three runs each, in the Study 2 rows of Table 22. The declared free-tier sweep was stopped after one model, superseded by Study 3 (Section 6), which runs four models from other developers on fresh firms.

A scoping defect, and what it touches. While preparing Study 3 we found that the scoped prompt for the wage rule never carried a generated firm's `actual_wage_system` export, which the rule needs. In Study 2 every wage pair was therefore sent without it. The 27 withheld wage pairs are unaffected, since their evidence was missing either way. The 27 decidable wage pairs (three firms, labelled `pass` or `fail`) were not decidable from what the models saw, and are excluded from every Study 2 metric. False-green rates and the Table 23 contrasts do not change. Detection and false-alarm figures do, and one reading in an earlier version of this paper was wrong: we attributed Gemini 3.1 Pro's near-miss false alarms (0.162) to caution about evidence that was present. They were requests for the missing actual-wage export, which was correct. The defect is fixed for Study 3.

Schema contrasts. On these firms no contrast survives correction (Table 23), consistent with firms that carry no proxy columns.

Generated firms are easier. With abstention offered, the three models report 0.021 to 0.071 of unassessable pairs as compliant, below their anchor-firm rates. Gemini 3.1 Pro and 3.8 Flash detect every violation and raise no false alarm on a conforming control, near-miss controls included, in three of the four schemas. Gemma 4 detects 0.346 of violations and, with abstention offered, raises 0.074 near-miss false alarms. The generated exports are clean, with no populated column next to a missing one, which is the gap Study 3 fills. Excluding the reverification pairs, whose violation labels assume no extension was filed (Section 9), changes no conclusion; no false green fell on them.

F Study 3 details and repeated-run tables

Repeats are close. Across the 24 model, study and schema cells, the three runs' false-green rates differ by at most 0.142, and by 0.06 or less in 21 of them. That bounds run-to-run variation on one host at temperature 0; it says nothing about variation across hosts, so it does not license the single runs of Section 7.

hris_roster_export.csv (first two workers; seed 1018, which withholds `i9_record_export`)

Plain

employeeNumber	visaType	jobTitle	hireDate	employmentStatus	…
SYN-1001	H-1B	Computer Systems Analysts	2024-11-05	Active	
SYN-1002	H-1B	Computer Systems Analysts	2022-02-03	Active	

Proxy

employeeNumber	visaType	hireDate	…	i9Status	onboardingCompletedDate	workAuthValidThrough	annualBaseSalary	payBandMidpoint
SYN-1001	H-1B	2024-11-05		Complete	2024-11-02	2027-04-29	118800	106800
SYN-1002	H-1B	2022-02-03		Complete	2022-01-30	2028-11-05	118800	108000

Figure 12. Study 3 firm 1018 as generated, rendered plain and with proxy columns. Both withhold the I-9 export, so every I-9 pair is unassessable in both; the proxy rendering adds five roster columns (shaded); every label and every other export is identical.

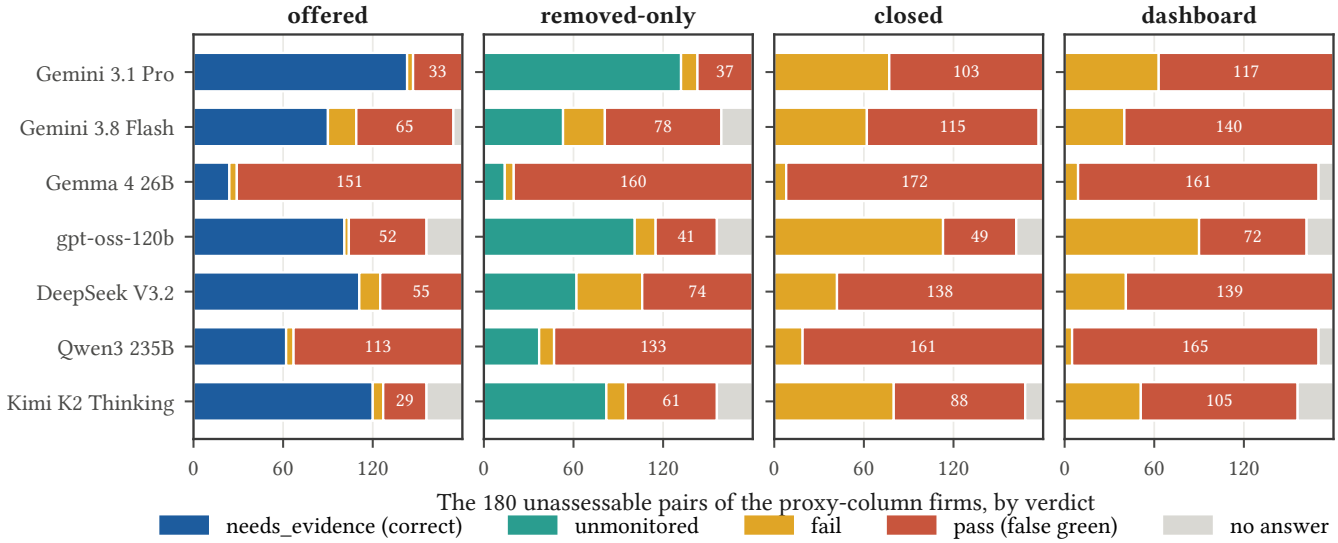


Figure 13. Study 3, proxy-column firms: where the 180 unassessable pairs go under each schema. Removing needs_evidence alone sends many pairs to unmonitored; removing every non-committal verdict sends them to fail or pass, in proportions that differ by model.

Table 21. Schema contrasts on plain and proxy-column firms (exploratory).

Change of schema	Plain firms (exploratory)	Proxy firms (declared test)
Abstention: offered → closed	+0.093 [+0.04, +0.16], p < 0.001 unadjusted	+0.259 [+0.20, +0.33], Holm p = 0.002
Definition and warning: closed → dashboard	+0.035 [+0.01, +0.06], p = 0.008 unadjusted	+0.084 [+0.04, +0.12], Holm p = 0.002

Pooled over the seven models with equal weight; same pairs, clusters and bootstrap draws as the declared analysis. The plain-firm column was computed after the results and is not part of any declared family.

G Interpretive choices of the reference checker

The checker of Section 4.3 records every choice the regulations leave open, with its source, in `labelcheck/INTERPRETATIONS.md`. The choices that matter most are these. Section 1 signed on the first day of employment is timely (8 C.F.R. 274a.2(b)(1)(i)(A); M-274 §3.0). Section 2 is due by the third business day after the first day, with weekends and federal holidays (with their observed dates) excluded; the M-274 does not define "business day". Reverification on the expiry date itself is timely. The required wage is the greater of the prevailing wage and the actual wage, compared per pay period rather than as a year-to-date average (20 C.F.R. 655.731(a), (c)(4)). None of the five proxy columns is treated as evidence. Section 4.3 gives the number of verdicts each boundary choice would change.

Across the checked firms, 70 Section 2 dates fall exactly on the deadline day; counting weekdays without excluding federal holidays would change 6 verdicts, and counting calendar days 55. The 1,176 distinct labels are 1,680 counting the proxy twins of Study 3, whose labels are identical by construction (`docs/labelcheck_results.json`).

H The same error in a rule engine

The defects below were found and fixed in the private repository of the engine that the rulebook comes from (Settle AI Inc.), between 2026-06-28 and 2026-07-14. Commit identifiers are given for the record; the repository is not public. In each case

Table 22. The three repeated-run models, both studies, four schemas. Mean of three runs, with the minimum and maximum in brackets.

Model	Study	Schema	False green	FG, answered only	Recall	False alarm, near-miss	Unassessable pairs answered pass / fail / needs_ev. / unmon.	Min answered
Gemini 3.1 Pro	1	offered	0.058 [0.058, 0.058]	0.058	0.433 [0.400, 0.450]	—	7 / 1 / 101 / 11	140/140
Gemini 3.1 Pro	1	removed-only	0.050 [0.025, 0.067]	0.050	0.550 [0.500, 0.600]	—	6 / 5 / 0 / 109	140/140
Gemini 3.1 Pro	1	closed	0.050 [0.033, 0.067]	0.050	0.950 [0.950, 0.950]	—	6 / 114 / 0 / 0	140/140
Gemini 3.1 Pro	1	dashboard	0.131 [0.125, 0.142]	0.131	0.917 [0.850, 0.950]	—	16 / 104 / 0 / 0	140/140
Gemini 3.1 Pro	2	offered	0.021 [0.016, 0.032]	0.021	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	3 / 0 / 123 / 0	225/252
Gemini 3.1 Pro	2	removed-only	0.056 [0.024, 0.071]	0.056	1.000 [1.000, 1.000]	0.025 [0.000, 0.037]	7 / 26 / 0 / 93	225/252
Gemini 3.1 Pro	2	closed	0.061 [0.056, 0.064]	0.061	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	8 / 118 / 0 / 0	225/252
Gemini 3.1 Pro	2	dashboard	0.061 [0.056, 0.064]	0.061	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	8 / 118 / 0 / 0	225/252
Gemini 3.8 Flash	1	offered	0.094 [0.092, 0.100]	0.094	0.500 [0.500, 0.500]	—	11 / 0 / 96 / 12	140/140
Gemini 3.8 Flash	1	removed-only	0.156 [0.125, 0.183]	0.160	0.783 [0.750, 0.850]	—	19 / 18 / 0 / 79	128/140
Gemini 3.8 Flash	1	closed	0.242 [0.208, 0.283]	0.245	0.767 [0.600, 0.850]	—	29 / 89 / 0 / 0	127/140
Gemini 3.8 Flash	1	dashboard	0.419 [0.375, 0.458]	0.433	0.833 [0.800, 0.850]	—	50 / 66 / 0 / 0	127/140
Gemini 3.8 Flash	2	offered	0.026 [0.016, 0.032]	0.027	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	3 / 4 / 117 / 0	219/252
Gemini 3.8 Flash	2	removed-only	0.037 [0.032, 0.048]	0.037	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	5 / 62 / 0 / 59	225/252
Gemini 3.8 Flash	2	closed	0.156 [0.127, 0.182]	0.156	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	20 / 106 / 0 / 0	225/252
Gemini 3.8 Flash	2	dashboard	0.167 [0.167, 0.167]	0.167	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	21 / 105 / 0 / 0	225/252
Gemma 4 26B	1	offered	0.153 [0.150, 0.158]	0.153	0.067 [0.050, 0.100]	—	18 / 0 / 76 / 26	140/140
Gemma 4 26B	1	removed-only	0.183 [0.175, 0.200]	0.220	0.067 [0.050, 0.100]	—	22 / 0 / 0 / 79	116/140
Gemma 4 26B	1	closed	0.600 [0.525, 0.667]	0.682	0.400 [0.400, 0.400]	—	72 / 33 / 0 / 0	122/140
Gemma 4 26B	1	dashboard	0.589 [0.567, 0.617]	0.635	0.400 [0.400, 0.400]	—	71 / 41 / 0 / 0	128/140
Gemma 4 26B	2	offered	0.071 [0.071, 0.071]	0.071	0.346 [0.296, 0.370]	0.074 [0.074, 0.074]	9 / 0 / 116 / 1	225/252
Gemma 4 26B	2	removed-only	0.071 [0.071, 0.071]	0.071	0.370 [0.333, 0.407]	0.111 [0.111, 0.111]	9 / 0 / 0 / 117	225/252
Gemma 4 26B	2	closed	0.172 [0.167, 0.175]	0.189	0.444 [0.370, 0.518]	0.049 [0.037, 0.074]	22 / 93 / 0 / 0	208/252
Gemma 4 26B	2	dashboard	0.175 [0.175, 0.175]	0.175	0.370 [0.370, 0.370]	0.037 [0.037, 0.037]	22 / 104 / 0 / 0	225/252

Schemas: *offered* (four verdicts, evidence-anchored definitions); *removed-only* (offered minus `needs_evidence`); *closed* (`pass`/`fail` only, evidence-anchored definitions kept); *dashboard* (`pass`/`fail` only, `pass` redefined as "the obligation is satisfied", warning dropped). The anchor subset has 120 unassessable pairs and no controls; Study 2 rates are pooled over its six firms (126 unassessable pairs). Min answered: the fewest pairs answered in any of the three runs. The verdict columns show every off-schema answer: none occurred on an unassessable pair in any cell.

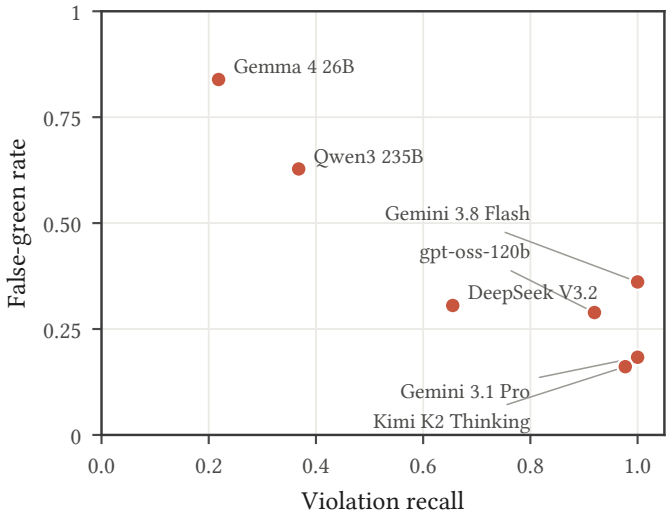


Figure 14. Study 3, proxy-column firms, offered schema: violation recall against false-green rate, one point per model. Higher recall goes with fewer false greens (Spearman −0.55), but no high-recall model is near zero.

Table 23. Paired schema contrasts, repeated-run models.

Model	Study	Change of schema	Unassessable pairs	Change in FGR [95% CI]	Holm p
Gemini 3.1 Pro	1	offered ⟶ removed-only	120	-0.008 [-0.04, +0.01]	1.000
Gemini 3.1 Pro	1	offered ⟶ closed	120	-0.008 [-0.09, +0.05]	1.000
Gemini 3.1 Pro	1	closed ⟶ dashboard	120	+0.081 [+0.02, +0.17]	0.036 **
Gemini 3.1 Pro	1	offered ⟶ dashboard	120	+0.072 [-0.04, +0.19]	1.000
Gemini 3.1 Pro	2	offered ⟶ removed-only	126	+0.034 [+0.00, +0.11]	1.000
Gemini 3.1 Pro	2	offered ⟶ closed	126	+0.040 [+0.00, +0.10]	1.000
Gemini 3.1 Pro	2	closed ⟶ dashboard	126	+0.000 [+0.00, +0.00]	1.000
Gemini 3.1 Pro	2	offered ⟶ dashboard	126	+0.040 [+0.00, +0.10]	1.000
Gemini 3.8 Flash	1	offered ⟶ removed-only	108	+0.077 [-0.01, +0.20]	1.000
Gemini 3.8 Flash	1	offered ⟶ closed	115	+0.136 [+0.04, +0.24]	0.019 **
Gemini 3.8 Flash	1	closed ⟶ dashboard	103	+0.168 [+0.10, +0.24]	0.012 **
Gemini 3.8 Flash	1	offered ⟶ dashboard	108	+0.324 [+0.17, +0.48]	0.012 **
Gemini 3.8 Flash	2	offered ⟶ removed-only	120	+0.003 [+0.00, +0.01]	1.000
Gemini 3.8 Flash	2	offered ⟶ closed	120	+0.133 [+0.00, +0.33]	0.960
Gemini 3.8 Flash	2	closed ⟶ dashboard	126	+0.011 [-0.01, +0.04]	1.000
Gemini 3.8 Flash	2	offered ⟶ dashboard	120	+0.142 [+0.00, +0.37]	1.000
Gemma 4 26B	1	offered ⟶ removed-only	97	+0.086 [+0.01, +0.20]	0.012 **
Gemma 4 26B	1	offered ⟶ closed	102	+0.542 [+0.38, +0.70]	0.012 **
Gemma 4 26B	1	closed ⟶ dashboard	102	-0.049 [-0.16, +0.04]	1.000
Gemma 4 26B	1	offered ⟶ dashboard	108	+0.509 [+0.34, +0.67]	0.012 **
Gemma 4 26B	2	offered ⟶ removed-only	126	+0.000 [+0.00, +0.00]	1.000
Gemma 4 26B	2	offered ⟶ closed	109	+0.110 [+0.01, +0.25]	0.320
Gemma 4 26B	2	closed ⟶ dashboard	109	+0.000 [+0.00, +0.00]	1.000
Gemma 4 26B	2	offered ⟶ dashboard	126	+0.103 [+0.02, +0.24]	0.102

Each unassessable pair's outcome is its share of `pass` over the three runs. Pairs answered in every run of both conditions; obligations resampled (2,000 draws, firm by obligation in Study 2); two-sided p-values Holm-corrected across all 24 contrasts; ** marks p < 0.05.

the engine returned a verdict that the available record could not support.

Because the repository is private, the logic of each defect is restated below as pseudocode (paraphrased, not copied), so that the claim can be checked against the description.

```
# silent skip (I-9 Section 2; reverification)
before: for worker in workers:
            record = i9_records.get(worker)
            if record is None or record.section2_date is None: continue   # worker leaves monitoring
            emit(check(record))
after:      if record is None or record.section2_date is None:
                emit(needs_evidence(worker, missing="Section 2 date")); continue
```

Table 24. Study 3, false-green rate by model, firm rendering and schema.

Model	Arm	FGR offered	removed-only	closed	dashboard	Recall, offered	False alarm, offered	Min answered
Gemini 3.1 Pro	plain	0.106	0.189	0.206	0.294	1.000	0.000	504/504
Gemini 3.1 Pro	proxy	0.183	0.206	0.572	0.650	1.000	0.000	504/504
Gemini 3.8 Flash	plain	0.161	0.217	0.350	0.361	1.000	0.000	498/504
Gemini 3.8 Flash	proxy	0.361	0.433	0.639	0.778	1.000	0.000	471/504
Gemma 4 26B	plain	0.339	0.333	0.406	0.411	0.230	0.084	498/504
Gemma 4 26B	proxy	0.839	0.889	0.956	0.894	0.218	0.093	494/504
gpt-oss-120b	plain	0.133	0.239	0.228	0.228	0.954	0.017	474/504
gpt-oss-120b	proxy	0.289	0.228	0.272	0.400	0.920	0.030	468/504
DeepSeek V3.2	plain	0.111	0.211	0.189	0.267	0.529	0.131	504/504
DeepSeek V3.2	proxy	0.306	0.411	0.767	0.772	0.655	0.127	504/504
Qwen3 235B	plain	0.317	0.350	0.367	0.394	0.345	0.194	489/504
Qwen3 235B	proxy	0.628	0.739	0.894	0.917	0.368	0.232	494/504
Kimi K2 Thinking	plain	0.156	0.161	0.261	0.233	0.966	0.000	474/504
Kimi K2 Thinking	proxy	0.161	0.339	0.489	0.583	0.977	0.034	456/504

Twelve fresh firms (504 pairs, 180 unassessable) per arm, one run per cell. *plain*: generator v2; *proxy*: the same firms with the five proxy columns (generator v3). False alarm is any non-`pass` verdict on one of the 237 conforming controls.

```
# salary of record as wages paid
before: verdict = PASS if roster.salary >= lca.required_wage else FAIL
after:  verdict = NEEDS_EVIDENCE unless payroll.periods cover the span; salary alone never passes

# greater-of rule without the actual-wage limb
before: required = lca.prevailing_wage; verdict = PASS if paid >= required else FAIL
after:  if actual_wage_benchmark is None and paid >= lca.prevailing_wage: verdict = NEEDS_EVIDENCE
        else: required = max(lca.prevailing_wage, actual_wage_benchmark); ...

# fabricated provenance
before: cohort.pw_source = "OES"; cohort.pw_doc_status = "ok"
after:  cohort.pw_source = mapped_column or "unknown"   # "unknown" is read as an evidence request
```

The engine's rules were written with AI coding assistance, and the defects were found by AI-assisted review (Codex), so the table is not evidence about human engineers. It shows that a system built to fill a dashboard acquires these substitutions easily, and that each fix is the same: return `needs_evidence` when the record that decides the obligation is absent.

I AI pilot runs of the expert protocol

The expert protocol is a workbook of 50 items and 8 statements (`docs/expert_review_v3/`, built by `scripts/build_expert_kit_v3.py`). Its 50 items are:

- 20 unassessable pairs on proxy-column firms;
- 10 unassessable pairs on plain firms;
- 20 decidable violations and controls, some on the legal boundary.

Each item shows the employer's records inline, and the reviewer answers blind to the construction label. No human expert has completed it.

What was run. Two AI systems completed it as a pilot, at the authors' request:

- Claude (Anthropic) completed version 2. Its own answers flagged five defects in the workbook, all fixed in version 3: identifiers repeated across firms, two items with identical records, no stated rule on status extensions, no holiday calendar, and no wage item that could fail.
- Codex (OpenAI) completed version 3.

Both are scored against their own version's labels (`scripts/score_ai_pilot.py`, `docs/ai_pilot_review/results.json`).

Results. Each agreed with all 50 labels:

- every proxy-column item was answered `needs_evidence`;
- every planted violation was answered `fail`;
- every control was answered `pass`.

On the evidence statements, both strongly disagreed that an I-9 status flag, an onboarding date or a future authorisation end date shows I-9 timeliness, and both strongly disagreed that an LCA's wage shows the wages paid. Both strongly agreed

Table 25. Study 3, the declared tests.

Estimate	P1 abstention	P2 definition and warning	P3 proxy columns, dashboard	P4 proxy columns, offered
Pooled, 7 models (primary)	+0.259 [+0.20, +0.33] p=0.002 **	+0.084 [+0.04, +0.12] p=0.002 **	+0.427 [+0.29, +0.56] p=0.002 **	+0.217 [+0.15, +0.29] p=0.002 **
Google, 3 models	+0.258 [+0.16, +0.36] p=0.004 **	+0.069 [+0.01, +0.13] p=0.021 **	+0.440 [+0.27, +0.61] p=0.004 **	+0.265 [+0.18, +0.35] p=0.004 **
Other developers, 4 models	+0.260 [+0.18, +0.34] p=0.004 **	+0.095 [+0.04, +0.14] p=0.004 **	+0.417 [+0.29, +0.53] p=0.004 **	+0.181 [+0.12, +0.26] p=0.004 **
Gemini 3.1 Pro	+0.389 [+0.24, +0.55] p=0.014 **	+0.078 [+0.00, +0.17] p=0.432	+0.356 [+0.22, +0.50] p=0.014 **	+0.078 [+0.01, +0.17] p=0.432
Gemini 3.8 Flash	+0.269 [+0.10, +0.46] p=0.014 **	+0.136 [+0.00, +0.27] p=0.432	+0.417 [+0.18, +0.64] p=0.015 **	+0.218 [+0.07, +0.40] p=0.033 **
Gemma 4 26B	+0.117 [+0.00, +0.25] p=0.432	-0.006 [-0.02, +0.00] p=1.000	+0.547 [+0.34, +0.72] p=0.014 **	+0.500 [+0.29, +0.69] p=0.014 **
gpt-oss-120b	-0.020 [-0.12, +0.07] p=1.000	+0.154 [+0.07, +0.23] p=0.014 **	+0.174 [+0.08, +0.25] p=0.014 **	+0.160 [+0.05, +0.30] p=0.015 **
DeepSeek V3.2	+0.461 [+0.28, +0.62] p=0.014 **	+0.006 [-0.06, +0.07] p=1.000	+0.506 [+0.35, +0.66] p=0.014 **	+0.194 [+0.06, +0.36] p=0.015 **
Qwen3 235B	+0.267 [+0.12, +0.40] p=0.014 **	+0.035 [-0.04, +0.11] p=1.000	+0.545 [+0.32, +0.74] p=0.014 **	+0.311 [+0.19, +0.45] p=0.014 **
Kimi K2 Thinking	+0.333 [+0.20, +0.49] p=0.014 **	+0.187 [+0.06, +0.31] p=0.070	+0.444 [+0.25, +0.63] p=0.015 **	+0.058 [-0.03, +0.17] p=1.000

Difference in false-green rate on unassessable pairs answered in both cells, [95% CI] from an obligation-cluster bootstrap (18 firm-by-rule clusters, 2,000 draws), Holm-adjusted p. Holm families as declared: the four pooled tests; the 28 per-model tests; the eight group tests. ** marks adjusted p < 0.05. Pooled estimates weight each model equally.

Table 26. Per-model effects of the declared tests of Studies 4–7.

Test	Gemini 3.1 Pro	Gemini 3.8 Flash	Gemma 4 26B	DeepSeek V3.2	GLM-5.2
H1 Names	+0.021	+0.115	+0.612	+0.091	-0.091
H2 Meaning	+0.021	+0.115	+0.758	+0.121	+0.115
H3 Direction (`fail`)	+0.000	+0.000	+0.076	+0.076	+0.051
H4 Abstaining verdicts	+0.519	+0.444	+0.188	+0.564	+0.725
H5 Bare definitions	-0.087	+0.001	-0.027	+0.045	+0.068
H6 Warning	+0.166	+0.114	-0.003	+0.059	-0.005
H7 Evidence-first prompt	+0.021	+0.067	+0.133	+0.106	+0.088
R1 Pay fields, offered	+0.000	+0.496	+0.182	+0.748	+0.382
R2 Pay fields, dashboard	+0.325	+0.000	+0.000	+0.092	+0.000
R3 Dashboard – offered	+0.583	+1.000	+0.175	+0.783	+1.000
W1 Pay Rate alone	+0.000	+0.160	+0.182	+0.775	+0.000
W2 Pay type and schedule	+0.000	+0.000	+0.175	+0.158	+0.000

Each model's effect on the false-green rate (H3: on `fail`), paired on pairs answered in both cells and pooled over firms; descriptive, computed after the results. Tests in Tables 5 and 6 and Section 6.8.

Table 27. Study 9 Part B by model.

Model	Unsupported pass: prompt	Unsupported pass: extraction	Correct: prompt	Correct: extraction	Follow-up: prompt	Follow-up: extraction
Gemini 3.1 Pro	0.000	0.000	1.000	1.000	0.649	0.649
Gemini 3.8 Flash	0.000	0.000	1.000	0.977	0.649	0.653
Gemma 4 26B	0.229	0.007	0.572	0.951	0.507	0.667
DeepSeek V3.2	0.108	0.002	0.573	0.961	0.682	0.660
GLM-5.2	0.002	0.004	0.946	0.944	0.658	0.663

Study 9 Part B, eighteen packets. Correct: correct decisions on decidable pairs. Follow-up: share of all pairs answered `fail`, abstaining or unanswered. After the solver correction described in Section 6.10.

Table 28. Study 13 by model.

Model	Unsupported pass: prompt	Unsupported pass: extraction	Correct: prompt	Correct: extraction	Extraction – parser, correct [95% CI]
Gemini 3.1 Pro	0.000	0.000	1.000	1.000	+0.323 [+0.191, +0.451]
Gemini 3.8 Flash	0.000	0.000	1.000	1.000	+0.323 [+0.191, +0.451]
Gemma 4 26B	0.392	0.043	0.408	0.935	+0.259 [+0.143, +0.368]
DeepSeek V3.2	0.222	0.016	0.321	0.921	+0.244 [+0.117, +0.376]
GLM-5.2	0.000	0.000	0.979	1.000	+0.323 [+0.191, +0.451]

Study 13 replication, held-out layouts, eighteen packets; intervals are packet bootstrap per model.

that a missing record should be reported as needing evidence. Claude disagreed, and Codex strongly disagreed, that a salary of record shows the required wage was paid and that an HR status field can mark an item compliant for triage.

What this shows. These runs are not expert validation:

- **Not blind to the study.** Both systems worked with the authors on this project, and Codex had reviewed earlier drafts of the paper, so neither was blind to the evidence standard.
- **Correlated errors.** Systems of the same kind as the models under test may share their errors.

The runs show that the protocol can be completed as designed, and that a careful reader of the standard reproduces the labels. Whether practitioners accept that standard remains open (§9).

Table 29. Study 14 by model: correct decisions and the declared safety criterion.

Model	System	Unsupported pass	Correct decisions	Excess unsupported pass over the parser [95% CI]	Meets 0.01 margin
Gemini 3.1 Pro	Extraction + solver	0.000	0.999	-0.003 [-0.007, +0.000]	yes
Gemini 3.1 Pro	Hybrid	0.003	0.992	+0.000 [+0.000, +0.000]	yes
Gemini 3.1 Pro	Code generation	0.000	0.983	-0.003 [-0.007, +0.000]	yes
Gemini 3.8 Flash	Extraction + solver	0.000	0.994	-0.003 [-0.007, +0.000]	yes
Gemini 3.8 Flash	Hybrid	0.003	0.988	+0.000 [+0.000, +0.000]	yes
Gemini 3.8 Flash	Code generation	0.000	0.967	-0.003 [-0.007, +0.000]	yes
Gemma 4 26B	Extraction + solver	0.047	0.704	+0.045 [+0.033, +0.057]	no
Gemma 4 26B	Hybrid	0.050	0.796	+0.047 [+0.034, +0.061]	no
Gemma 4 26B	Code generation	0.014	0.430	+0.011 [+0.006, +0.018]	no
DeepSeek V3.2	Extraction + solver	0.015	0.769	+0.012 [+0.006, +0.017]	no
DeepSeek V3.2	Hybrid	0.015	0.802	+0.013 [+0.007, +0.018]	no
DeepSeek V3.2	Code generation	0.004	0.431	+0.001 [-0.005, +0.005]	yes
GLM-5.2	Extraction + solver	0.001	0.998	-0.002 [-0.007, +0.001]	yes
GLM-5.2	Hybrid	0.003	0.992	+0.000 [+0.000, +0.000]	yes
GLM-5.2	Code generation	0.000	0.792	-0.003 [-0.007, +0.000]	yes

Study 14, thirty contents. The declared criterion: the content-bootstrap upper bound of the excess over the rule-based parser (unsupported pass 0.003) is below 0.01.

Table 30. Study 15 by model.

Model	Prompt: unsupported pass	Prompt: correct	Extraction: unsupported pass [95% CI]	Extraction: correct	Code generation: unsupported pass (95% upper bound)	Code generation: correct
Gemini 3.1 Pro	0.000	0.990	0.000 [0.000, 0.000]	1.000	0.000 (≤0.0025 per pair)	1.000
Gemini 3.8 Flash	0.000	0.990	0.000 [0.000, 0.000]	1.000	0.000 (≤0.0025 per pair)	1.000
Gemma 4 26B	0.314	0.505	0.007 [0.003, 0.011] (fails)	0.841	0.000 (≤0.0025 per pair)	0.962
DeepSeek V3.2	0.167	0.509	0.005 [0.001, 0.011] (fails)	0.986	0.000 (≤0.0025 per pair)	0.998
GLM-5.2	0.000	0.974	0.001 [0.000, 0.003]	0.993	0.000 (≤0.0025 per pair)	0.900

Study 15, corrected run, fifty packets. The declared criterion is a packet-bootstrap upper 95% bound below 0.01. With no event observed, the exact one-sided 95% bound is shown instead, per pair; it assumes independent pairs, and per packet it is 0.058.

Defect	What the code did	Fix
Silent skip, I-9 Section 2 and reverification	An evaluator skipped an employee whose record lacked the deciding dates, so the employee left monitoring and the dashboard stayed green	Emit a visible `needs_evidence` finding (commits `16de553c`, `407eb438`; an internal review note calls it "a verified production defect")
Salary of record as wages paid	The material-change wage check returned pass or fail from the HR salary of record	Retired: a salary of record "proves nothing about wages paid"; it never passes alone (commits `1b181e00`, `4b2605ec`)
Required wage without the actual-wage limb	The greater-of rule passed on the prevailing wage alone when the employer's actual-wage benchmark was unknown	The would-be pass becomes `needs_evidence` (commit `55c5e6fb`)
Fabricated provenance	Wage cohorts were stamped with a prevailing-wage source and an "ok" document status that no input supplied	Stored as unknown, so the rules ask for the record (commit `c47a945b`)